



LINK400TM

Product Information

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Section 1

Introduction

This section presents an overview of LINK400. It also lists the supported protocols, interface, products and external message format.

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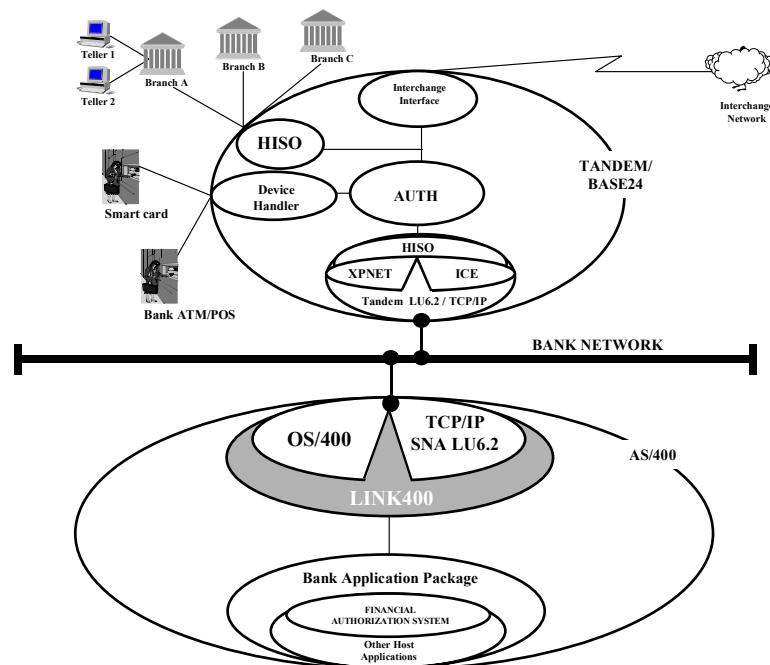
Overview

What is LINK400?

LINK400 is an application interface developed by ACI to run on the IBM AS/400. It links an application running on an AS/400 to the Host Interface of BASE24 using the LU6.2 or TCP/IP communications protocol. LINK400 also takes care of the message formatting and routing if required.

LINK400 consists of four main modules. These are:

- The Receiver, which receives messages from BASE24.
- The Sender, which sends messages to BASE24.
- The Formatter, which breaks up ISO 8583 messages into the host application format.
- The Reformatter, which formats the host application message back to the ISO 8583 format.



Two Flavors of LINK400

LINK400 is available in either LU6.2 or TCP/IP version.

The **LINK400-LU62** version caters for customers that need SNA LU6.2 connectivity between BASE24 and their back-end host application. LINK400-LU62 can be implemented over SDLC, Ethernet, X.25 and TCP/IP. It is written using the RPG/400 programming language and uses CPI-C (Common Programming Interface Communications) communications features on the AS/400.

The **LINK400-TCP/IP** version caters for customers who wish to deploy pure TCP/IP connectivity. LINK400-TCP/IP runs faster than the LU6.2 version since it takes advantage of the communications speed offered by the TCP/IP protocol. It is written using the ILE C/400 programming language and uses the AS/400 sockets API that is compatible with the Berkeley Software Distributions (BSD) 4.3 sockets.

Throughout this document, the two versions will be simply referred to as LINK400. Any differences will be particularly referred to either LINK400-TCP/IP or LINK400-LU62.

Supported Communications Protocols, Products, Interfaces, External Message and Standards

Communications Protocols

LINK400 currently supports TCP/IP and LU6.2 protocols. Those who wish to use pure TCP/IP protocol can opt for LINK400-TCP/IP. Otherwise, LINK400-LU62 can be used.

BASE24 Interfaces

LINK400 supports the BASE24 Host and BIC Interfaces. Changes may be required to LINK400 if it will be set to handle network management messages from the BASE24 BIC Interface.

Products

LINK400 can support BASE24-atm and BASE24-pos products. In addition to these, it can support other products that are ISO-based with possibly some changes to LINK400.

External Message Format

LINK400 supports the ISO 8583 standard.

Issuer & Acquirer Processing

LINK400 supports both issuer and acquirer processing. This means that LINK400 can deliver messages that originate from either BASE24 or the AS/400 host application.

High Transaction Volume Support

LINK400 can be implemented in a high transaction volume environment. Aside from the inherent speed offered by the TCP/IP protocol (i.e. in the case of LINK400-TCP/IP), the two versions of LINK400 can accept multiple conversation requests from BASE24. This effectively allows LINK400 to process more transactions concurrently and speed up the transaction turnaround time.

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Section 2

Components

LINK400 comprises a number of processes and entities to fulfill its task.

The communications programs are perhaps the basic and most important processes of LINK400. These programs will be referred to as the Receiver and Sender throughout this document.

The formatter programs, on the other hand, are used to format messages from ISO to host format and vice versa. These are optional processes and are used when the host application does not support ISO messages. The formatter programs will be referred to as the Formatter and Reformatter throughout this document.

Additional processes that are inherent to LINK400-TCP/IP will also be described in this section. These are the Receiver SAF File Monitor, Sender SAF File Monitor and the Settlement Initiator.

Other important entities that will be introduced are data queues, logfiles, SAF files and debug files.

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Individual Processes

Communications Programs

Receiver

The Receiver is responsible for receiving messages from BASE24. It does this by accepting communications conversation request from BASE24. Once the Receiver binds and establishes the conversation, the Receiver waits until it receives messages from BASE24.

In LINK400-LU62, the Receiver is invoked as soon as the Tandem operator initiates the conversation request from BASE24. Likewise, the Receiver stops running as soon as the operator ends the conversation. Note that the Receiver will automatically start only if the appropriate LINK400 subsystem is active.

In LINK400-TCP/IP, however, the Receiver is started and ended by the AS/400 operator. Once the Receiver starts running, it will remain active and ready to receive subsequent or pending communications conversation request(s) by the Tandem operator.

Additionally, the LINK400-TCP/IP Receiver acts as a server in a typical TCP/IP environment. BASE24 must therefore be configured to run as a client.

	LINK400-TCP/IP	LINK400-LU62
Receiver Program Name	TCPRCV	B24RCV

Sender

The Sender is responsible for sending messages to BASE24. It does this by accepting communications conversation request from BASE24. Once the Sender binds and establishes the conversation, the Sender waits until it receives messages from the Reformatter and/or host application.

In LINK400-LU62, the Sender is automatically started and ended by the Receiver.

In LINK400-TCP/IP, however, the Sender always starts together with the Receiver. The Sender also establishes the conversation request in a similar way as the Receiver. It acts as a server as well.

	LINK400-TCP/IP	LINK400-LU62
Sender Program Name	TCPSND	B24SND

Formatter Programs

Formatter

The Formatter is an optional LINK400 process. It is used when the host application does not support ISO messages and therefore requires LINK400 to convert the ISO message to the host format before sending to the host application. The Formatter is automatically started as soon as the appropriate LINK400 subsystem is started.

In LINK400-TCP/IP, multiple Formatter jobs can be run for load balancing processing. This will be explained later in the *LINK400-TCP/IP Section*.

The Formatter will rely mainly on the format expected by the host application. The Formatter, therefore, is always customized for each customer.

	LINK400-TCP/IP	LINK400-LU62
Formatter Program Name	FORMATTER	ISOFMTR

Reformatter

The Reformatter is another optional LINK400 process that runs in parallel with the Formatter. It is used to convert the host message back to ISO format. The Reformatter is automatically started as soon as the appropriate LINK400 subsystem is started.

In LINK400-TCP/IP, multiple Reformatter jobs can also be run for load balancing processing. This will also be explained in the *LINK400-TCP/IP Section*.

Like the Formatter, the Reformatter is normally customized for each customer.

	LINK400-TCP/IP	LINK400-LU62
Reformatter Program Name	REFORMAT	ISODFMTR

Store-and-Forward (SAF) File Monitor Programs

The SAF File Monitor programs are only available in the LINK400-TCP/IP version and runs together with the Receiver and Sender.

When LINK400-TCP/IP fails to deliver the message among its processes (i.e. Receiver, Sender, Formatter & Reformatter), BASE24 or the host application, it writes the message to the SAF file. The SAF File Monitor programs check these files and ensure that if messages are present in the SAF files, these messages are written

back to the appropriate LINK400 data queue to be redelivered to the intended process.

Receiver and Sender SAF File Monitor Program Name	QSAFMON
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Receiver SAF File Monitor

The Receiver SAF File Monitor constantly checks in 30-second interval if messages are written to the Receiver/Formatter SAF file. If there are, then these messages are moved back to the appropriate Receiver data queue until the SAF file is empty.

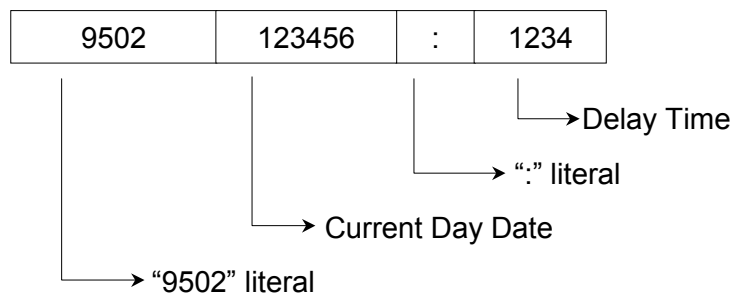
Sender SAF File Monitor

The Sender SAF File Monitor constantly checks in 30-second interval if messages are written to the Sender/Reformatter SAF file. If there are, then these messages are moved back to the appropriate Sender data queue until the SAF file is empty.

Settlement Initiator

This sleep program wakes up at a defined settlement time to send cutover messages to the Formatter and Reformatter. The Formatter and Reformatter use these messages to mark that they need to close the current day logfiles, tag the next day logfiles as the current day logfiles and lastly, open the new next day logfiles.

The cutover message format is:



Cutover Message Components		
Component	Length	Remarks
"9502" literal	4 bytes	This is the cutover message header
Current Day Date	6 bytes	Date in yymmdd format
:	1 byte	
Delay Time	4 bytes	Delay time in seconds

The delay time is used by the Formatter and Reformatter to determine the time interval before it continues to process subsequent messages. This prevents the Formatter and Reformatter from reading cutover messages that are intended for other Formatter and Reformatter jobs (in case of multiple Formatter and Reformatter jobs processing).

The delay value will always be zero when only pair of Formatter and Reformatter is running. However, if there are two or more Formatter and Reformatter jobs, the Settlement Initiator computes the delay time (i.e. in 30-second increment) for each of the cutover message that it sends out.

To illustrate, if there are two jobs for each Formatter and Reformatter, the first pair of cutover messages will have a 30-second delay while the second pair will have zero. If there are three jobs, the first pair of cutover messages will have a 60-second delay, the second pair will have 30-second delay and the last will have zero and so on.

After the Settlement Initiator has sent all cutover messages, it sleeps for another six minutes to allow all the Formatter and Reformatter jobs to complete the settlement processing. It then awakens to purge old Formatter and Reformatter logfiles that exceeded the retention period before finally going back to sleep and wait until the next settlement time.

Settlement Initiator Program Name	SETCTLCL
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Data Queues and Files

Data Queue

Data queues provide a fast means of asynchronous communications between LINK400 processes – one LINK400 process can run independently regardless of the status of the other processes. This enables LINK400 to continue to receive and process messages from BASE24 even while the response message is still on its way back to BASE24. The actual number of data queues used in each LINK400 process differs in both LINK400-TCP/IP and LINK400-LU62.

For LINK400-TCP/IP:

Process	Data Queue	Usage
Receiver	Receiver Data Queue	Store incoming ISO messages. The Formatter or the host application must read from this data queue.
Sender	Sender Data Queue	Store outgoing ISO messages. The Reformatter or the host application must

Components

Process	Data Queue	Usage
		write to this data queue.
Receiver SAF File Monitor	Receiver Data Queue	Store messages taken from the Receiver/Formatter SAF file. This is the same data queue that the Receiver writes to.
Sender SAF File Monitor	Sender Data Queue	Store messages taken from the Sender/Reformatter SAF file. This is the same data queue that the Sender reads from.
Formatter	Receiving Data Queue	Store incoming ISO messages. This is the same data queue that the Receiver writes to.
	Sending Data Queue	Store outgoing host messages. The host application must read from this data queue.
	Buffer Data Queue	Store a copy of the parsed ISO message.
Reformatter	Receiving Data Queue	Store incoming host messages. The host application must write to this data queue.
	Sending Data Queue	Outgoing ISO messages. This must correspond to one Sender data queue.
	Buffer Data Queue	Retrieve original ISO messages previously written by the Formatter. This is the same buffer data queue that the Formatter uses.
Settlement Initiator	Formatter Receiving Data Queue	Send settlement messages. This must be the same data queue that the Formatter reads from.
	Reformatter Receiving Data Queue	Send settlement messages. This must be the same data queue that the Reformatter reads from.

For LINK400-LU62:

Process	Data Queue	Usage
Receiver	Receiver Data Queue	Store incoming ISO messages. The Formatter or the host application must read from this data queue.
Sender	Sender Data Queue	Store outgoing ISO messages. The Reformatter or the host application must write to this data queue.
Formatter	Receiving Data Queue	Store incoming ISO messages. This is the same data queue that the Receiver writes to.
	Sending Data Queue – ATM transactions	Store outgoing host messages. The host application that is responsible for authorizing ATM transactions must read

Process	Data Queue	Usage
		from this data queue.
	Sending Data Queue – POS transactions	Store outgoing host messages. The host application that is responsible for authorizing POS transactions must read from this data queue.
Reformatter	Receiving Data Queue	Store incoming host messages. The two host applications (i.e. one for ATM and another for POS) must write to this data queue.
	Sending Data Queue – ATM transactions	Outgoing ISO messages. This must correspond to the Sender data queue for ATM transactions.
	Sending Data Queue – POS transactions	Outgoing ISO messages. This must correspond to the Sender data queue for POS transactions.

Logfiles

LINK400 can write incoming and outgoing messages to different AS/400 database files.

The following logfiles are used by LINK400-TCP/IP:

Filename	Description
TCPINLOG	Receiver Incoming ISO Messages Logfile
TCPOTLOG	Sender Outgoing ISO Messages Logfile
FFIyymmdd	Formatter Incoming ISO Messages Logfile
FFOyymmdd	Formatter Outgoing Host Messages Logfile
FMTEXLOG	Formatter Exception Messages Logfile
FRIyymmdd	Reformatter Incoming ISO Messages Logfile
FROyymmdd	Reformatter Outgoing Host Messages Logfile
RFMTEXLOG	Reformatter Exception Messages Logfile

The following logfiles are used by LINK400-LU62:

Filename	Description
B24INLOG	Receiver Incoming ISO Messages Logfile
B24OTLOG	Sender Outgoing ISO Messages Logfile

Store-and-Forward (SAF) Files

Store and forward (SAF) files are only used in LINK400-TCP/IP. When LINK400-TCP/IP fails to send the message to the intended process (e.g. Formatter, Sender, host

Components

application or BASE24), it writes the message to the SAF file. The SAF File Monitor processes will later pick up these messages and send back to the appropriate process.

The following SAF files are used by LINK400-TCP/IP:

Filename	Description
SAFI	Receiver and Formatter SAF File
SAFO	Sender and Reformatter SAF File

Debug Files

Like SAF files, debug files are only used in LINK400-TCP/IP. All LINK400-TCP/IP processes but the Settlement Initiator can be configured to write debug information into these files:

Process	Debug Filename	Remarks
Receiver	Profile Name + “_..._R”	The 10-byte filename is created by using the profile name that the Receiver and Sender belong to. Succeeding spaces are replaced with “_” and the last two bytes with “_R” or “_S.”
Sender	Profile Name + “_..._S”	
Receiver SAF File Monitor	DEBUGSAFI	
Sender SAF File Monitor	DEBUGSAFO	
Formatter	DEBUGFMnn	“nn” refers to the job number of the Formatter and Reformatter
Reformatter	DEBUGRFMnn	

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Section 3

LINK400-TCP/IP

This section details the features, processing and run configuration of LINK400-TCP/IP.

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LINK400-TCP/IP Features

Basic Network Management Message Support

LINK400-TCP/IP can handle network management messages. The Receiver can be configured to reply to logon, logoff and/or echo messages.

This feature is used to verify that a valid communications link between BASE24 and LINK400 exists and that messages can be sent through. To check whether the host application is available for authorization, LINK400-TCP/IP must be configured to send the network management message to the host application. However, the source code of the Formatter and Reformatter must be changed accordingly.

LINK400-TCP/IP also provides an option for the AS/400 operator to send a logon and/or logoff message to BASE24.

Multiple BASE24 Host Interfaces/Message Routers & AS/400 Host Applications Support

LINK400-TCP/IP supports multiple conversations from BASE24. This is achieved by creating *profiles*.

A *profile* defines a single group of the Receiver and Sender that directly accepts communications conversation request from a pair of BASE24 sending and receiving stations. Additionally, it collectively defines

- Interface parameters to identify the logical port ID and data queues that BASE24 stations, Receiver, Sender, Formatter, Reformatter and/or host application will use to establish the communications link and read/write messages for asynchronous communication.
- Run parameters to instruct the Receiver and/or Sender to log messages and audit traces, translate messages from ASCII to EBCDIC and vice versa, respond to logon, logoff and/or echo messages and clear individual data queues on startup. There is also a run parameter to attach sender data queue details to the Receiver's outgoing ISO message (see *Attach Sender Data Queue Details* in this section).

NOTE: The logical port ID must be unique in all *profiles* and the Sender and Receiver data queue must not be the same. However, the same pair of data queues may or may not be defined similarly in the other *profiles*. This will be explained later in this section.

Up to ten *profiles* can be defined and set to run concurrently.

Since each pair of BASE24 sending and receiving stations can be tied up to one Host Interface, LINK400-TCP/IP effectively supports up to ten Host Interfaces. Additionally, it can link a *profile* to one host application. However, multiple host applications will not be supported when the formatter programs are used since the latter currently support a single host application.

Character Set Translation

LINK400-TCP/IP can perform character set translation. It can do an ASCII to EBCDIC translation on incoming messages (i.e. messages coming from BASE24) and EBCDIC to ASCII on outgoing messages (i.e. messages going to BASE24). Translation is done in the Receiver and Sender.

Communications Recovery

LINK400-TCP/IP can recover when the communications link to BASE24 has been disrupted. This means that when the Receiver and/or Sender detect(s) that the associated BASE24 stations have been stopped at any point after the initial connection (i.e. conversation), the Receiver and Sender will wait until the affected BASE24 stations are restarted (by the Tandem operator). This ensures that the Receiver and Sender are always ready to accept conversations and process messages as soon as BASE24 gets ready to send messages.

The Receiver and Sender has its own *autorecovery flag* which must be set to YES prior to starting LINK400. It is not advised to set these flags to NO in the production environment.

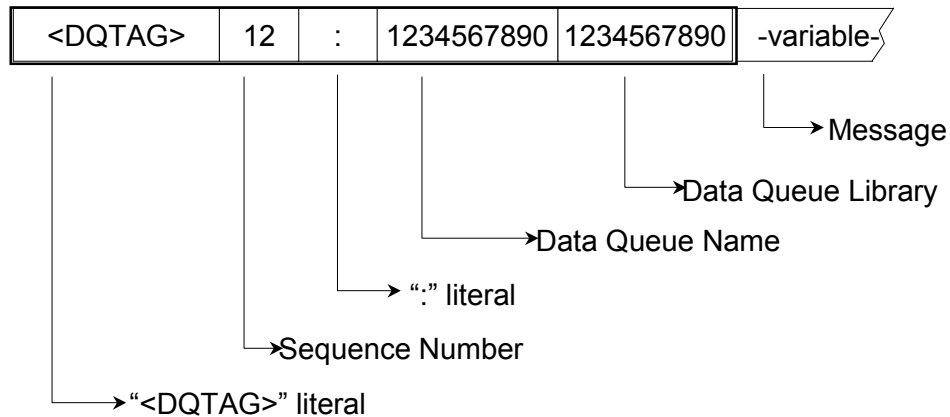
Attach Sender Data Queue Details

The Receiver can be configured to attach its partner Sender data queue details to the message prior to sending to the Formatter and/or the host application. This is normally used in a multiple BASE24 Host Interface conversation where all Receiver groups write to the same data queue.

When the Formatter sees the sender data queue details in the message, it reattaches these details in the host message before sending to the host application. The host application, in turn, must also attach these details to its response message so the Reformatter will determine if it needs to override its own sending data queue to route the message.

If the Formatter is not used, the host application will no longer need to reattach the sender data queue details to the response message. The host application will just use these details to determine which Sender it will return the message to.

The format of the data queue tag is:



Data Queue Tag Components		
Component	Length	Remarks
<DQTAG>	7 bytes	Literal searched for by the Formatter, Reformatter and/or the host application to identify that data queue details have been tagged to the message.
Sequence Number	2 bytes	Job Sequence Number of the <i>profile</i>
:	1 byte	
Data Queue Name	10 bytes	The partner Sender data queue. If the data queue name is less than ten, the remaining bytes are padded with blanks.
Data Queue Library	10 bytes	The library where the data queue is located. If the library name is less than ten, the remaining bytes are padded with blanks.
Message	Variable length	The actual ISO or host message

Formatters Load Balancing Processing

Up to ten Formatter and Reformatter jobs can run concurrently. This is particularly useful when the Receiver and Sender process transactions faster than a single Formatter and Reformatter job can handle.

This may happen on the following scenarios:

- Multiple conversations (i.e. Multiple Receiver and Sender) can cause messages to come in at once and sent to the Formatter.

- Message logging facility is used in the Formatter and Reformatter. File write operation still gives off some overhead time and may be felt during extremely peak transaction periods. It is therefore recommended that message logging facility is used only as required.

All jobs of the Formatter and Reformatter will use the same interface and run parameters that are defined in the *Formatter Parameters Maintenance* and similarly share the same resources (e.g. logfiles, data area, etc.). However, if audit tracing is used, separate debug files will be created for each job.

Message Logging and Logfiles Settlement Processing

Messages that are processed within the Receiver, Sender, Formatter and Reformatter can be logged to individual logfiles.

Message logging facility on the Receiver and Sender processes is intended for testing purposes and may not be a requirement in the production environment. If messages are needed to be logged in production, the Formatter and Reformatter offers a more flexible way of logging messages to database files.

The Formatter and Reformatter can log ISO and/or host messages to separate daily log files according to the message capture date (i.e. P-17). Note that LINK400-TCP/IP rejects the message if the capture date is not equal to the LINK400-TCP/IP current day or next day dates. Using daily logfiles has the following advantages:

- It is easier to track and view all daily transactions.
- Old logfiles can be easily maintained according to the specified retention period.

New logfiles are automatically created to store next day transactions while old logfiles that exceeded the retention period are deleted. These are all done by the Settlement Initiator and its processing will be explained later in this section.

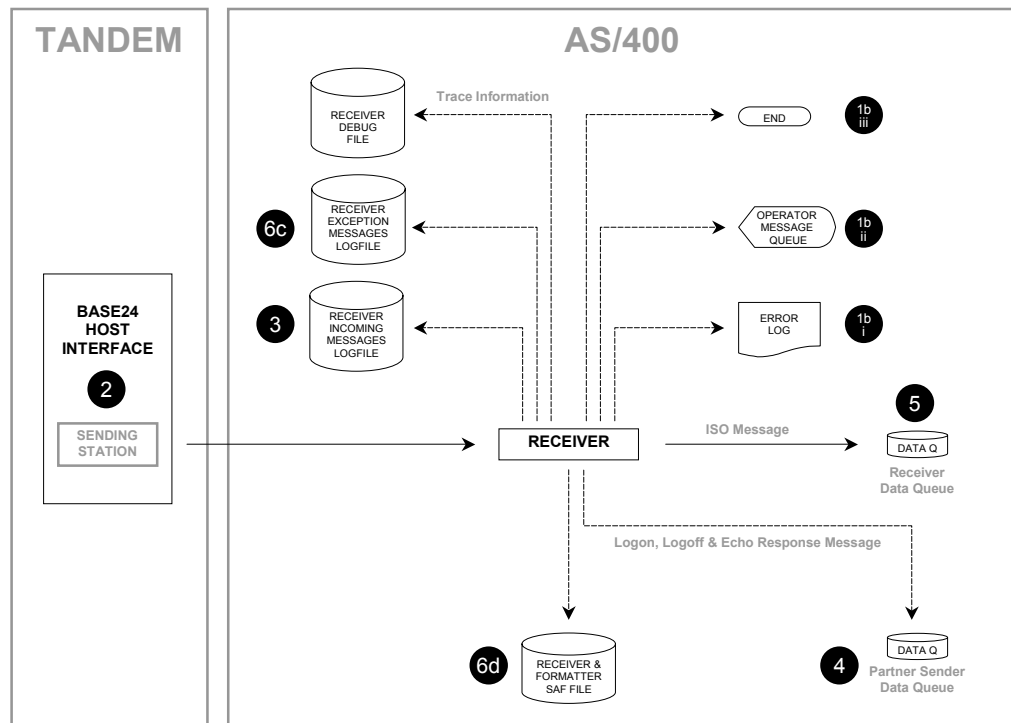
Audit Tracing

LINK400-TCP/IP provides a debugging facility that allows its processes (i.e. Receiver, Sender, SAF File Monitor, Formatter and Reformatter) to log the values of important program variables in separate debug files. The contents of these files can be viewed anytime.

Note that the debug file quickly increases in size and frequent write operations to the debug file considerably slow each process that uses this feature. This feature must only be used during testing and never in production.

Transaction Processing

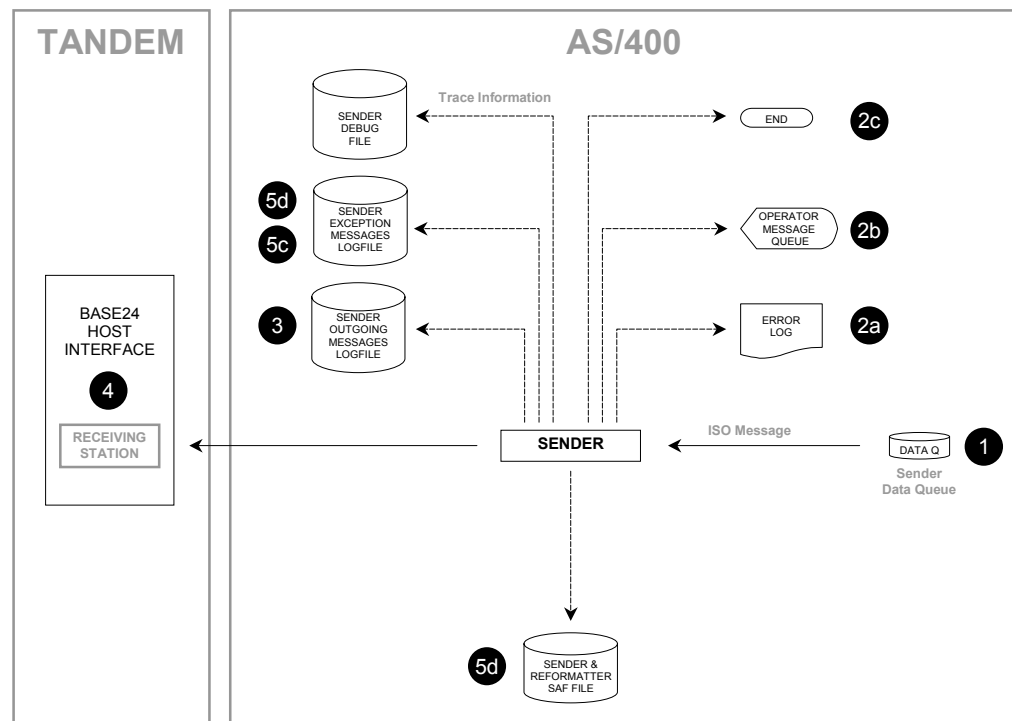
Receiver



1. The Receiver waits until BASE24 sends a message. If it detects an error in the communications link
 - a. If the *autorecover flag* is set to YES, it waits until the associated BASE24 station is restarted. The Receiver then resumes normal transaction processing (i.e. go back to Step #1).
 - b. Otherwise it performs abnormal processing
 - i. The Receiver writes the error message to a spoolfile.
 - ii. It sends a message to the LINK400 operator message queue to inform that the Receiver has ended.
 - iii. It ends immediately.
2. The Receiver receives a message from BASE24. If the *translate flag* is set to YES, it translates the original message's character set from ASCII to EBCDIC.
3. If the *log incoming messages flag* is set to YES, the Receiver writes the message to the incoming messages logfile.
4. If the message is a network management message
 - a. If this is a logon message and *respond to logon flag* is set to YES, the Receiver builds the response message and sends it to its partner Sender.
 - b. If this is a logoff message and *respond to logoff flag* is set to YES, the Receiver builds the response message and sends it to its partner Sender.

- c. If this is an echo message and *respond to echo flag* is set to YES, the Receiver builds the response message and sends it to its partner Sender.
5. Otherwise, it sends the message to the Formatter (or host application if the Formatter is not used). Additionally, if the *attach sender data queue details flag* is set to YES, it attaches the partner sender data queue details to the message.
6. If the send fails
 - a. If the message is not an ISO message (i.e. "ISO" literal is missing), the Receiver executes abnormal processing (see Step #1b).
 - b. If the message is a network management message, the Receiver drops the message.
 - c. If the message is a 0100, 0200, 0300, 0500, 0600, 0121, 0221, 0321, 0421, 0521 or 0621, the Receiver writes the message in the exception file. If this fails, it executes abnormal processing (see Step #1b).
 - d. If the message is a 0120, 0220, 0310, 0320, 0402, 0420, 0520, 0620, 0110, 0130, 0210, 0230, 0330, 0412, 0430, 0510, 0530, 0610 or 0630, the Receiver writes the message to the SAF file. If the write fails, it executes abnormal processing (see Step #1b).
 - e. Otherwise, it just drops the message.
7. It waits until it receives another message from BASE24 (i.e. go back to Step #1).

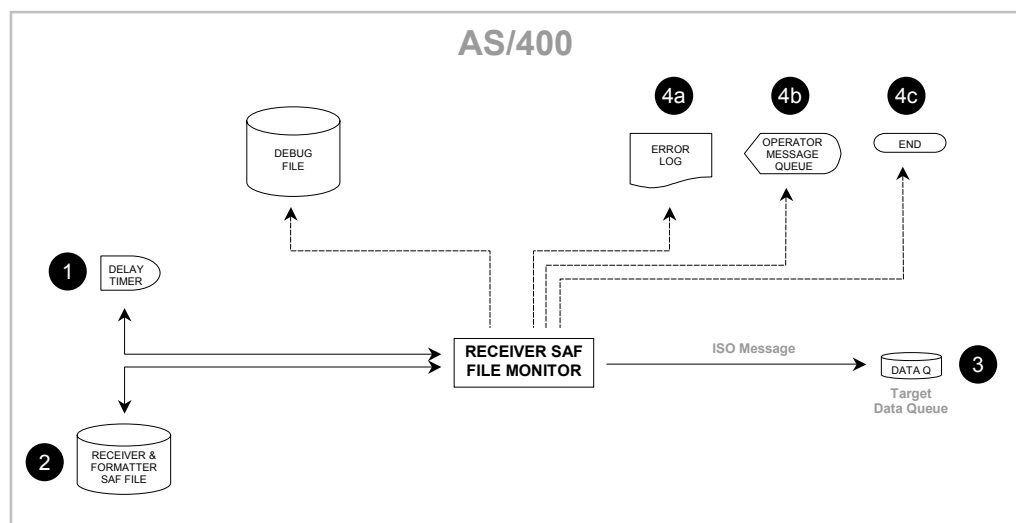
Sender



1. The Sender waits until it receives a message from the Reformatter (or host application if the Reformatter is not used).

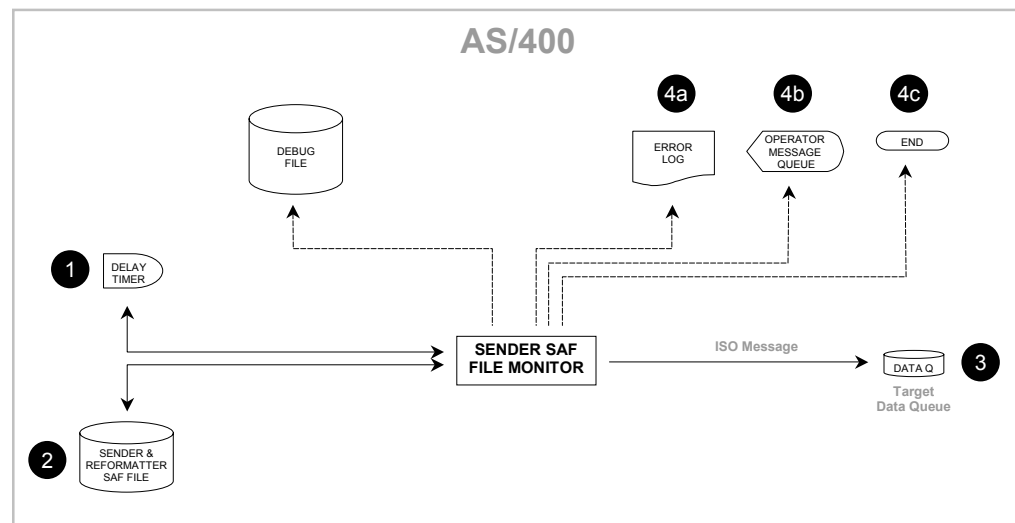
2. If the Sender detects an error in accessing the data queue, it executes the following abnormal processing
 - a. The Sender writes the error message to a spoolfile.
 - b. It sends a message to the LINK400 operator message queue to inform that the Sender has ended.
 - c. It ends immediately.
3. If the *log outgoing messages flag* is set to YES, the Sender writes the message to the outgoing messages logfile.
4. The Sender sends the message to BASE24. If the *translate flag* is set to YES, the Sender translates the original message's character set from EBCDIC to ASCII.
5. If the send fails (due to communications link error)
 - a. If the message is not an ISO message (i.e. "ISO" literal is missing), the Sender executes abnormal processing (see Step #2).
 - b. If the message is a network management message, the Sender drops the message.
 - c. If the message is a 0100, 0200, 0300, 0500, 0600, 0121, 0221, 0321, 0421, 0521 or 0621, the Sender writes the message in the exception file. If this fails, it executes abnormal processing (see Step #2).
 - d. If the message is a 0120, 0220, 0310, 0320, 0402, 0420, 0520, 0620, 0110, 0130, 0210, 0230, 0330, 0412, 0430, 0510, 0530, 0610 or 0630, the Sender writes the message to the exception file and SAF file. If this fails, it executes abnormal processing (see Step #2).
 - e. Otherwise, it just drops the message.
 - f. If the *autorecover flag* is set to YES
 - i. The Sender waits until the associated BASE24 station is restarted.
 - ii. It resumes normal transaction processing (i.e. go back to Step #1).
 - g. Otherwise, it executes abnormal processing (see Step #2).
6. It waits until it receives another message from the Reformatter or host application (i.e. go back to Step #1).

Receiver SAF File Monitor



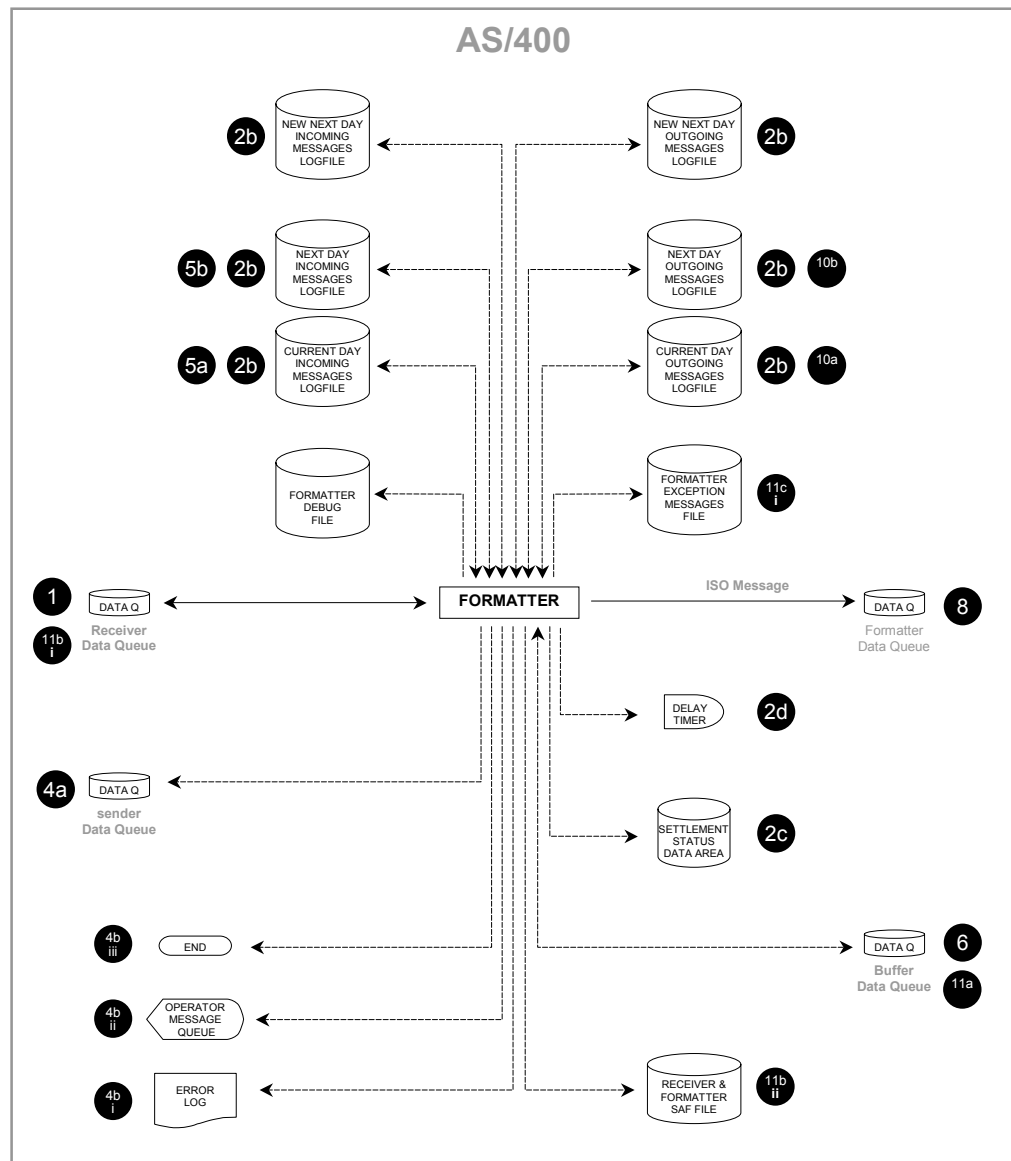
1. The Receiver SAF File Monitor wakes up (after 30 seconds) and checks if there are messages in the Receiver & Formatter SAF file. If the file is empty, the Receiver SAF File Monitor returns to sleep.
2. Otherwise, the Receiver SAF File Monitor reads a message from the file. It then parses the message record to get the ISO message and the target data queue that the message must be written to. The target data queue is particularly important as it allows LINK400 to return the message to the correct *profile* that owns the message.
3. The Receiver SAF File Monitor writes the message back to the specified data queue.
4. If the write fails, the Receiver SAF File Monitor executes abnormal processing
 - a. It writes the error message to a spoolfile.
 - b. It sends a message to the LINK400 operator message queue to inform that the Receiver SAF File Monitor has ended.
 - c. It ends immediately.
5. The Receiver SAF File Monitor processes the next message until the Receiver and Formatter SAF File is empty.

Sender SAF File Monitor



The processing is similar to the Receiver SAF File Monitor. The only differences are the use of the Sender & Reformatter SAF file and that it writes to a different debug file.

Formatter

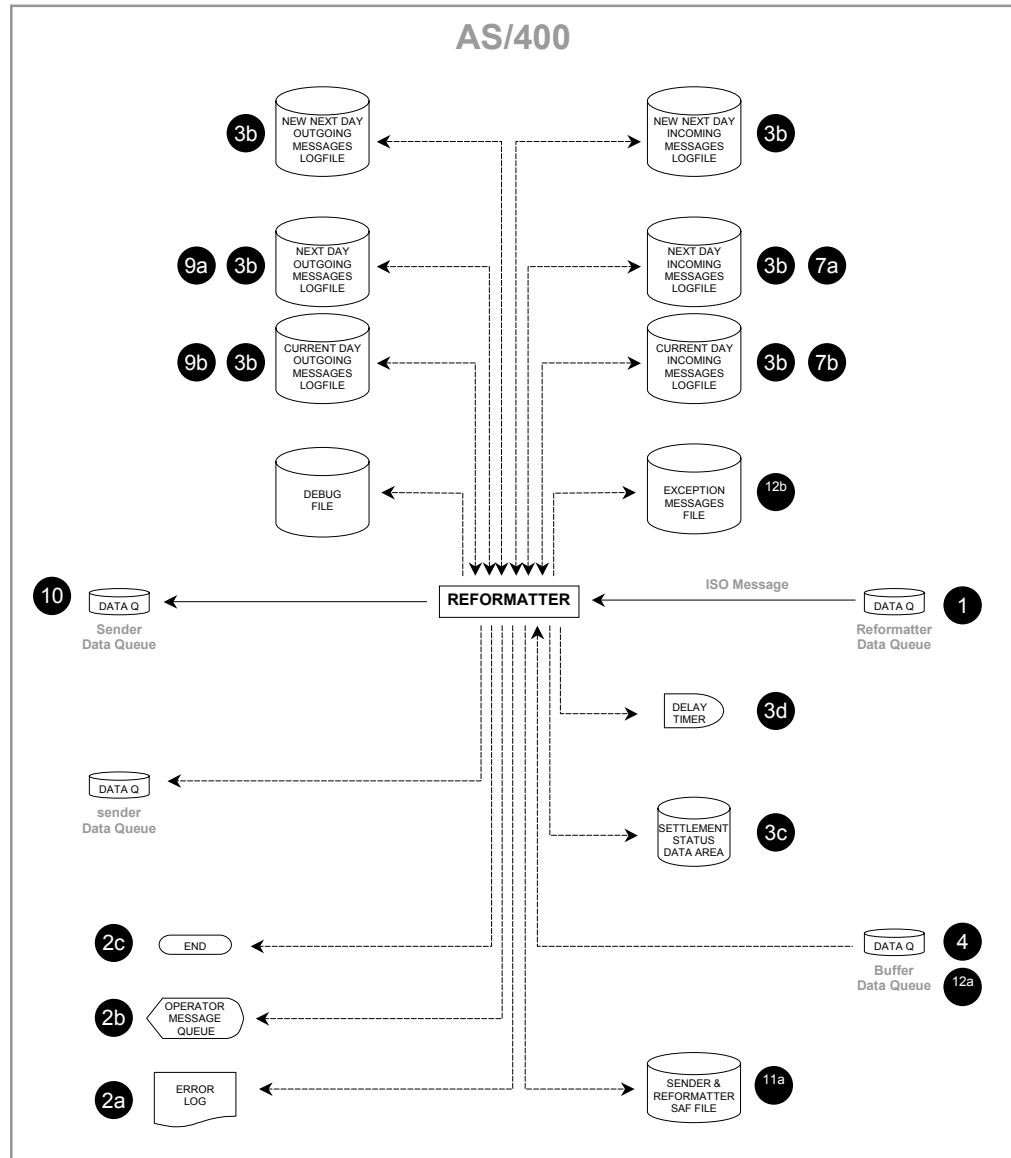


1. The Formatter waits until it receives a message from the Receiver and/or Settlement Initiator. If the message contains Sender data queue details, it parses the message to isolate the data queue details and the actual message.
2. If the message is a cutover message
 - a. If the Formatter already performed a cutover for the day (i.e. the message is a duplicate cutover message), it drops the message and waits for the next message (i.e. go back to Step #1).
 - b. Otherwise, it closes the Current day logfiles, marks the Next day logfiles as the Current day logfiles and opens the new Next day logfiles. Note that the

- Formatter opens incoming and outgoing logfiles only if the corresponding *log incoming messages* and/or *log outgoing messages flags* are set to YES.
- c. It signals that the settlement has completed.
 - d. The Formatter suspends processing according to the delay time specified in the cutover message.
 - e. The Formatter waits for the next message (i.e. go back to Step #1).
3. The Formatter parses the message to its individual primary and secondary bitmap fields.
 4. If the message capture date (P-17) is not equal to the Formatter's current day and next day dates
 - a. The Formatter rejects and sends the message back to the Sender. The Formatter routes the message to the Sender by using either its data queue information or the message's own Sender data queue details if there are any.
 - b. If the send fails
 - i. The Formatter writes the error message to a spoolfile.
 - ii. It sends a message to the LINK400 operator message queue to inform that the Formatter has ended.
 - iii. It ends immediately.
 5. Otherwise, if the *log incoming messages flag* is set to YES, the Formatter writes the message to its incoming messages logfile
 - a. If the message capture date (i.e. P-17) is equal to the Formatter's current day date, the Formatter writes the message to its current day logfile.
 - b. Otherwise, the Formatter writes the message to its next day logfile.
 6. The Formatter writes a copy of the (parsed) message to the buffer data queue (the Reformatter will later use this when rebuilding the response message). If the write to the buffer data queue fails, the Formatter executes abnormal processing (see Step #4b).
 7. The Formatter builds the host message. As there are different formatting requirements, this part will always be changed.
 8. The Formatter sends the host message to the host application. If Sender data queue details originally exist for this message, the Formatter attaches these details at the beginning of the host message.
 9. If the send fails, it executes abnormal processing (see Step #4b).
 10. Otherwise, if the *log outgoing messages flag* is set to YES, the Formatter writes the host message to its outgoing logfile.
 - a. If the message's Capture Date (i.e. P-17) is equal to the Formatter's current day date, the Formatter writes the message to its current day logfile.
 - b. Otherwise, the Formatter writes the message to its Next day logfile.
 11. If there are other unexpected errors while sending the message to the host application
 - a. Formatter deletes the copy of the message in the buffer data queue
 - b. If this is a 0220 or 0420 message
 - i. The Formatter rewrites the message back to originating data queue.
 - ii. If this fails, it writes the message to the SAF file. The Formatter includes the originating data queue details so when it is later picked up by the Receiver SAF File Monitor, the latter knows where it will later route the message.
 - iii. If this still fails, it executes abnormal processing (see Step #4b).

- c. Otherwise
 - i. The Formatter writes the message to its exception file.
 - ii. If this fails, it executes abnormal processing (see Step #4b).
12. The Formatter waits for the next message (i.e. go back to Step #1).

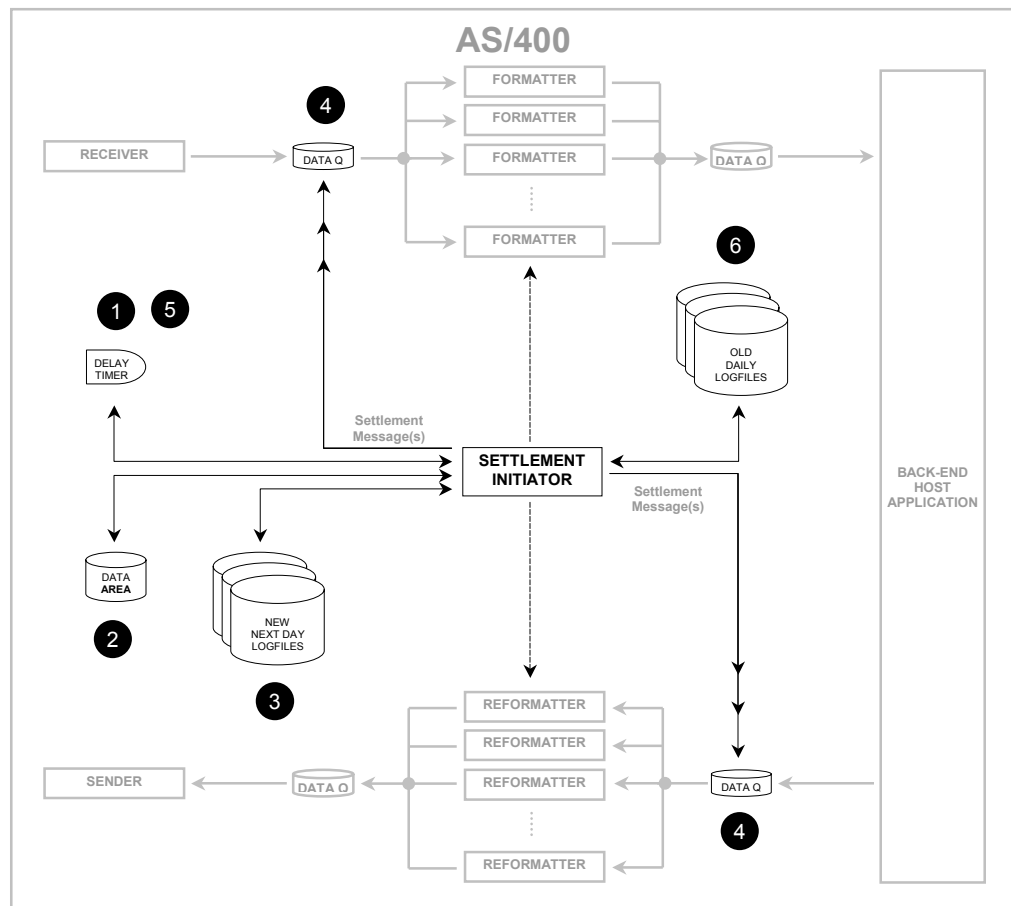
Reformatter



1. The Reformatter waits until it receives a message from the host application and/or Settlement Initiator. If the message contains Sender data queue details, it parses the message to isolate the data queue details and the actual message.

2. If an error is encountered while receiving the message, the Reformatter performs the following abnormal processing
 - a. It writes the error message to a spoolfile
 - b. It sends a message to the LINK400 operator message queue to inform that the Reformatter has ended
 - c. It ends immediately.
3. If the message is a cutover message
 - a. If the Reformatter already performed a cutover for the day (i.e. the message is a duplicate cutover message), it drops the message and waits for the next message to process (i.e. go back to Step #1).
 - b. Otherwise, it closes the current day logfiles, marks the next day logfiles as the current day logfiles and opens the new next day logfiles. Note that the Reformatter opens incoming and outgoing logfiles only if the corresponding *log incoming messages* and/or *log outgoing messages flags* are set to YES.
 - c. It signals that the settlement has completed.
 - d. The Reformatter suspends processing according to the delay time specified in the cutover message.
 - e. The Reformatter waits for the next message (i.e. go back to Step #1).
4. Otherwise, the Reformatter retrieves the copy of the message from the buffer data queue.
5. If an error is encountered while retrieving the message or the associated message does not exist, the Reformatter performs abnormal processing (see Step #2).
6. Otherwise, it parses the message to its individual primary and secondary bitmaps.
7. If the *log incoming messages flag* is set to YES, the Reformatter writes the host message to its incoming messages logfile
 - a. If the message capture date (i.e. P-17) is equal to the Reformatter's next day date, the Reformatter writes the message to its next day logfile.
 - b. Otherwise, the Reformatter writes the message to its current day logfile.
8. The Reformatter rebuilds the ISO message. As there are different formatting requirements, this part will always be changed.
9. If the *log outgoing messages flag* is set to YES, the Reformatter writes the ISO message to its outgoing logfile.
 - a. If the message capture date (i.e. P-17) is equal to the Reformatter's next day date, the Reformatter writes the message to its next day logfile.
 - b. Otherwise, the Reformatter writes the message to its current day logfile.
10. The Reformatter sends the ISO message to the Sender. If Sender data queue details exist for this message, the Reformatter uses this to correctly route the message. Otherwise, it uses its own data queue information.
11. If the send fails
 - a. The Reformatter writes the ISO message to the SAF file (the Reformatter includes the target data queue details to let the Sender SAF File Monitor determine where it will route the message).
 - b. If this fails, it executes abnormal processing (see Step #2).
12. If there are other unexpected errors
 - a. The Reformatter deletes the copy of the message in the buffer data queue
 - b. It writes the message to its exception file.
13. The Reformatter waits for the next message (i.e. go back to Step #1).

Settlement Initiator



1. Settlement Initiator wakes up at the specified settlement time.
2. It makes the Next day date as the new current day date and computes the new Next day date.
3. It creates the new Formatter and Reformatter Next day logfiles.
4. It sends cutover messages to the Formatter and Reformatter.
5. It sleeps for six minutes to allow all Formatter and Reformatter jobs to rollover its logfiles to the new dates.
6. The Settlement Initiator deletes old logfiles according to the retention period.
7. The Settlement Initiator goes back to sleep and waits for the next settlement time.

Run Configurations

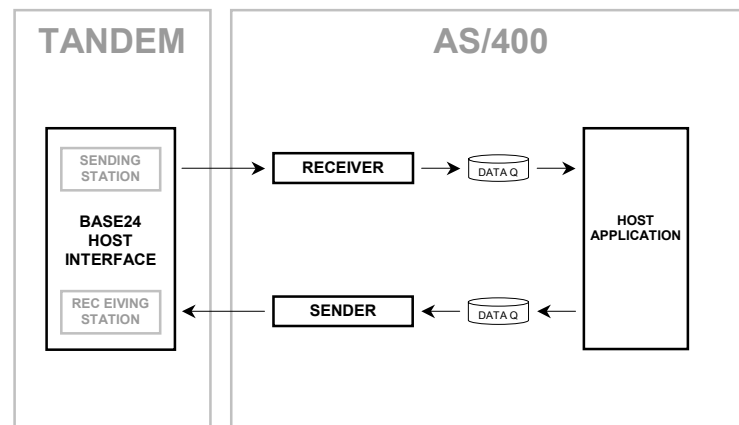
A normal LINK400-TCP/IP configuration consists of a Receiver, Sender, Formatter and Reformatter. While this configuration does the job of sending and receiving messages between BASE24 and the host application, LINK400-TCP/IP allows other configurations when any of the following scenarios are met:

- The back-end host can process ISO messages. Therefore, the formatter programs are no longer required.
- Multiple BASE24 stations and/or Host Interface are used to transmit messages to one or more back-end host application(s).
- The Formatter and Reformatter takes more processing time than the Receiver and Sender and therefore requires the former to run multiple jobs.

Communications Programs Used

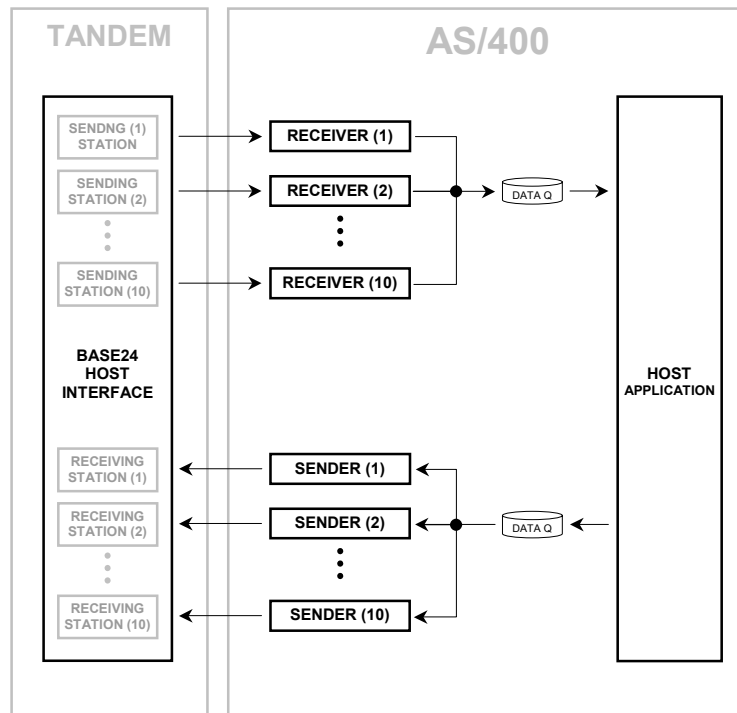
In the following configurations, it is assumed that the back-end host can process ISO messages. Hence, the formatter programs will no longer be required.

Base



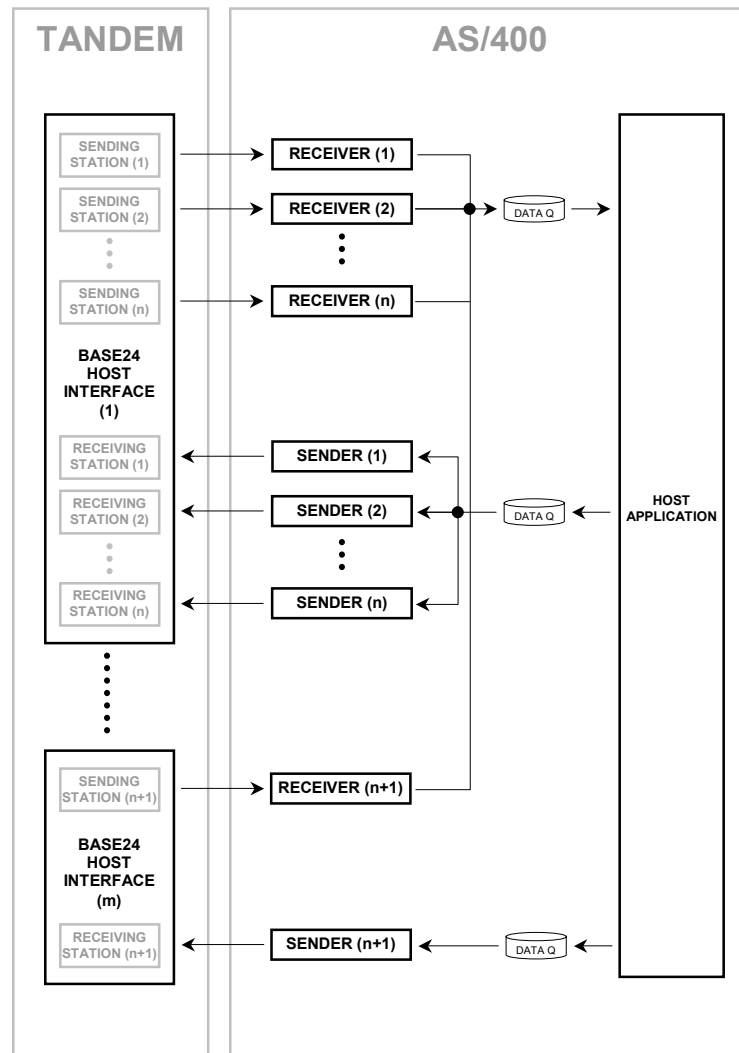
This is the simplest LINK400 configuration where one pair of Receiver and Sender (i.e. one *profile*) is tied up to a pair of BASE24 stations. This configuration can handle the normal daily transaction processing for most customers.

Multiple Message Routers



This configuration is similar to the base configuration except that more message routers exist to improve traffic throughput. This means that multiple BASE24 stations are used to send/receive messages to/from LINK400-TCP/IP. LINK400-TCP/IP can accept connection request from up to ten pairs of BASE24 sending and receiving stations by defining ten *profiles* accordingly. In this example, there is only one BASE24 Host Interface and host application so the same pair of data queues is used on all *profiles*.

Multiple Message Routers and Host Interfaces



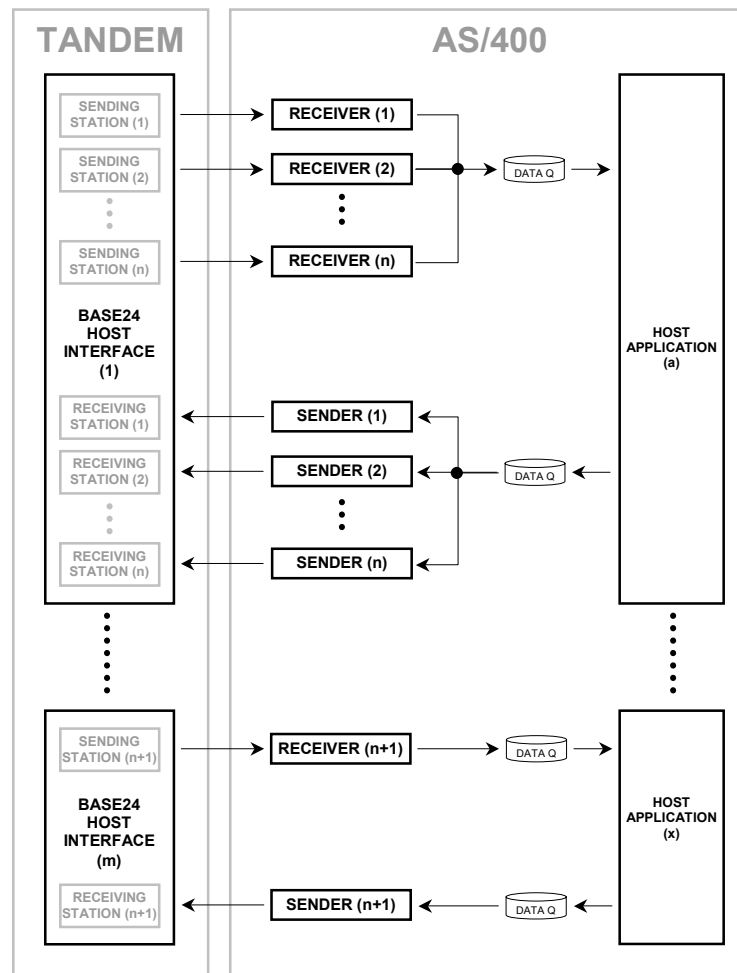
This is another typical Multiple Message Router Configuration with two or more Host Interfaces.

Messages must always be sent back to the originating BASE24 Host Interface. LINK400-TCP/IP facilitates this by

- Defining a unique data queue for each Sender. The number of Sender data queue used must correspond to the number of Host Interfaces.
- Configuring each (or all) Receiver(s) to attach the partner Sender data queue details to the message prior to sending the message to the host application. This will provide the information to the host application to determine the right Sender that the response message must be sent to.

The number of BASE24 Host Interface that LINK400-TCP/IP can support depends on the number of available *profiles*. If, for example, three *profiles* are already used, then LINK400-TCP/IP can support seven more Host Interfaces, assuming that only one *profile* will be used for each additional BASE24 Host Interface. However, two or more *profiles* can be linked to one Host Interface.

Multiple Message Routers, Host Interfaces and Host Applications



This configuration is used when a BASE24 Host Interface is tied up to a unique AS/400 host application. All Receiver and Sender jobs that are associated to one back-end host will use its own set of data queues. Hence, there is no need to instruct LINK400-TCP/IP to attach the partner Sender data queue details.

Again, the number of BASE24 Host Interface and host application that LINK400-TCP/IP can support depends on the number of available *profiles*.

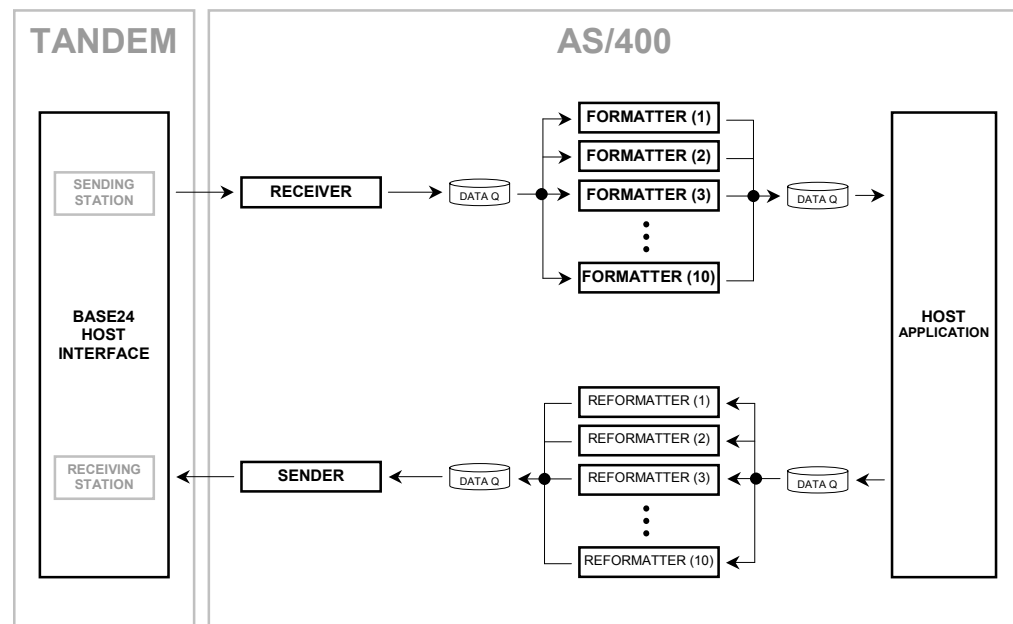
Communications and Formatters Programs Used

Unlike the communications programs, the formatter programs do not currently support the use of *profiles*. This limits the formatter programs to transmit messages between a lone source (i.e. communications programs) and destination (i.e. AS/400 host application). However, up to ten Formatter and Reformatter jobs can be set to run for load balancing purpose.

The right number of jobs depends on the transaction volume during peak transaction periods. It is recommended that the number is slowly increased until BASE24 does not timeout any message it sends to the host application. This ensures that no system resource is wasted in running unnecessary AS/400 jobs.

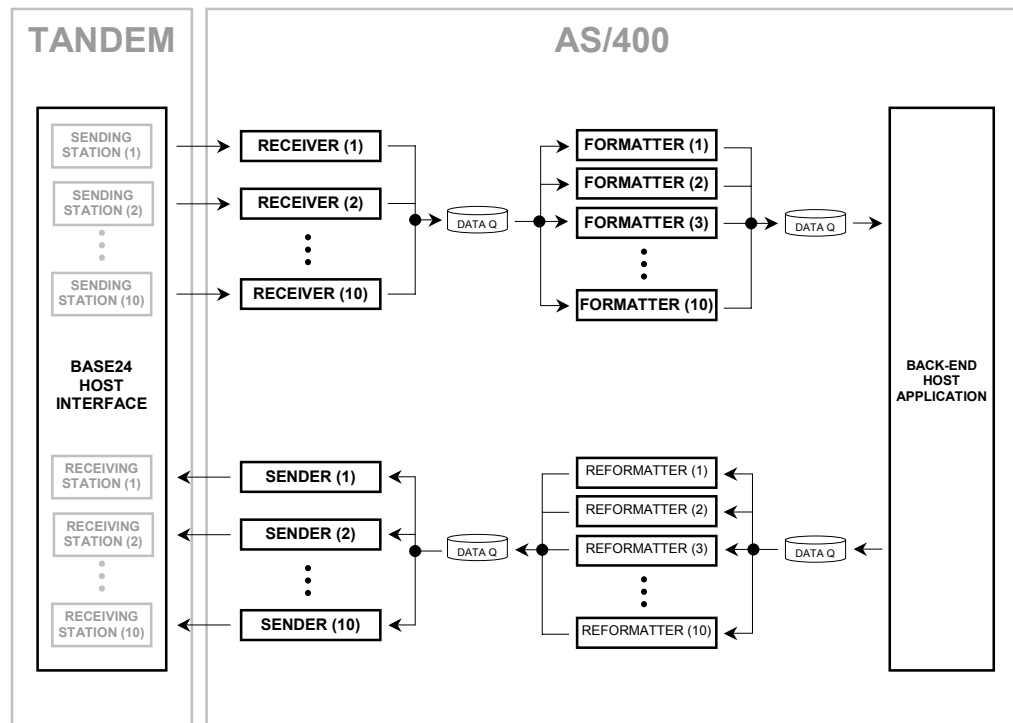
In the subsequent configurations, the back-end host does not process ISO messages and therefore requires LINK400-TCP/IP to handle the message formatting. The Formatter and Reformatter are therefore added in the configuration.

Base



This is the simplest and complete configuration involving a Receiver, Sender, Formatter and Reformatter. The Receiver and Sender directly exchanges messages to BASE24 while the Formatter and Reformatter take care of converting messages from ISO to host format and vice versa before passing to the host application.

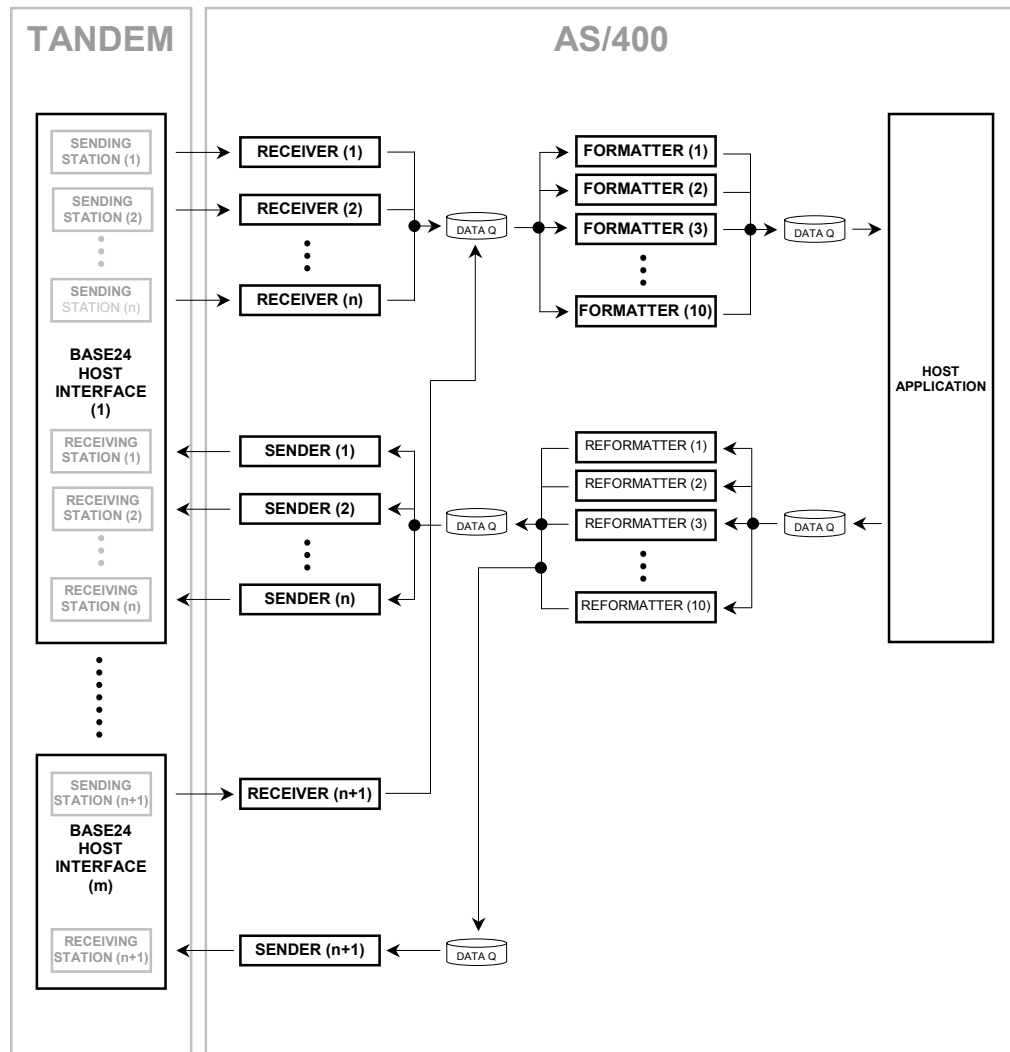
Multiple Message Routers with Formatters



This configuration shows the formatter programs in a typical multiple message routers environment involving a lone BASE24 Host Interface and a host application.

The multiple routers, are transparent to the Formatter and Reformatter since the formatter programs view the routers as one source. Also, remember that the number of Formatter and Reformatter jobs may not necessarily equal the number of message routers used.

Multiple Message Routers/Host Interfaces with Formatters



This configuration is almost identical to the multiple message routers configuration except that this involves multiple BASE24 Host Interfaces as well.

While the Formatter can only read/write messages from/to a pair of data queues, it is still possible for the Formatter to receive/send transactions originating from multiple BASE24 Host Interfaces. This is done by configuring the Receiver to attach the partner Sender data queue details to the ISO message.

Whenever the Formatter receives a message, it checks the message for sender data queue details. If there are, then it reattaches these details to the host message. The host application must include these details to the response message so when the Reformatter finally receives the response message, the Reformatter will determine which Sender the message will be routed to.

Section 4

LINK400-LU62

This section details the features and transaction processing that are inherent to the LINK400-LU62 version. While there are some LINK400-TCP/IP features that do not apply here, LINK400-LU62 also has its own unique features and processing. Nevertheless, LINK400-LU62 still fulfills the basic function of interfacing BASE24 and the AS/400 host application.

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LINK400-LU62 Features

Basic NMM Support

LINK400-LU62 can also handle basic logon, logoff and/or echo network management messages. The Receiver can be configured to reply or just pass these messages to the Formatter (or to the host application if the Formatter is not used). The Formatter, in turn, can continue to pass these messages to the host application or forward the messages back to the Sender. Again, this depends on the specified configuration. Additionally, LINK400-LU62 can be automatically send echo messages to BASE24 according to a specified time interval.

Like LINK400-TCP/IP, this feature is used to verify that a valid communications link between BASE24 and LINK400-LU62 exists and that messages can be sent through. To check whether the host application is available for authorization, LINK400-LU62 must be configured to send network management messages to the host application.

LINK400-LU62 also provides an option for the AS/400 operator to send a logon and/or logoff message to BASE24. Note that this feature is valid only when the BASE24 Host Interface is used. Some changes to LINK400-LU62 may be required when it needs to process network management messages from other ISO-based interfaces like the BIC Interface, for instance.

Multiple Message Routers, BASE24-atm/pos and Two Host Applications Support

With LINK400-LU62, two conversation groups can be established with BASE24. The two groups talk to BASE24-atm and BASE24-pos respectively. Each group also supports multiple Receiver and Sender conversation.

The maximum Receiver conversations depend on the BASE24 configuration as well as the AS/400 mode configuration. On the AS/400, there will be one job for each Receiver conversation.

The maximum Sender conversations, on the other hand, are limited to five per group but only one job will be active for all conversations. In addition, the sending conversations are split into primary and backup conversations. The backup conversations will be used only if all the primary conversations have failed.

Character Set Translation

LINK400-LU62 can also perform character set translation from EBCDIC to ASCII and vice versa.

Message Logging

ISO Messages that are processed within the Receiver and Sender can be logged to individual logfiles. This option is mainly intended for testing purposes and may not be required in the production environment.

SAF Extract to AS/400

In case of communications failure between the TANDEM and the AS/400, there is the ability to extract the entire BASE24 SAF file on to a tape, and restore it on to the AS/400. A selection by post date for the Extract is provided in case the SAF records include more than one day's data.

- The entire SAF file may be extracted. On the AS/400, the tape is inserted in the drive and a date is specified on the SAF extract/refresh screen. As the messages read from the tape are in ISO 8583 format, the messages will have to be passed on to the Formatter before a comparison can be made for the date.
- Before the messages are sent to the formatter, the letters "ISO" in the header of the messages will be changed to "ISX". This will indicate to the formatter that the messages are not coming from the TANDEM directly, but from a tape. The date entered on the screen will be appended to the end of the message.
- The Formatter will receive the messages. If it finds "ISX" in the header, it will take the post date at the end of the message and compare it to the post date in field P-17. If they are equal, the message will be processed and forwarded to the host application. Otherwise, the message will be dropped, and the next message will be processed.
- To process transactions for a different day, the same tape may be used but a different date must be specified.

File Transfer

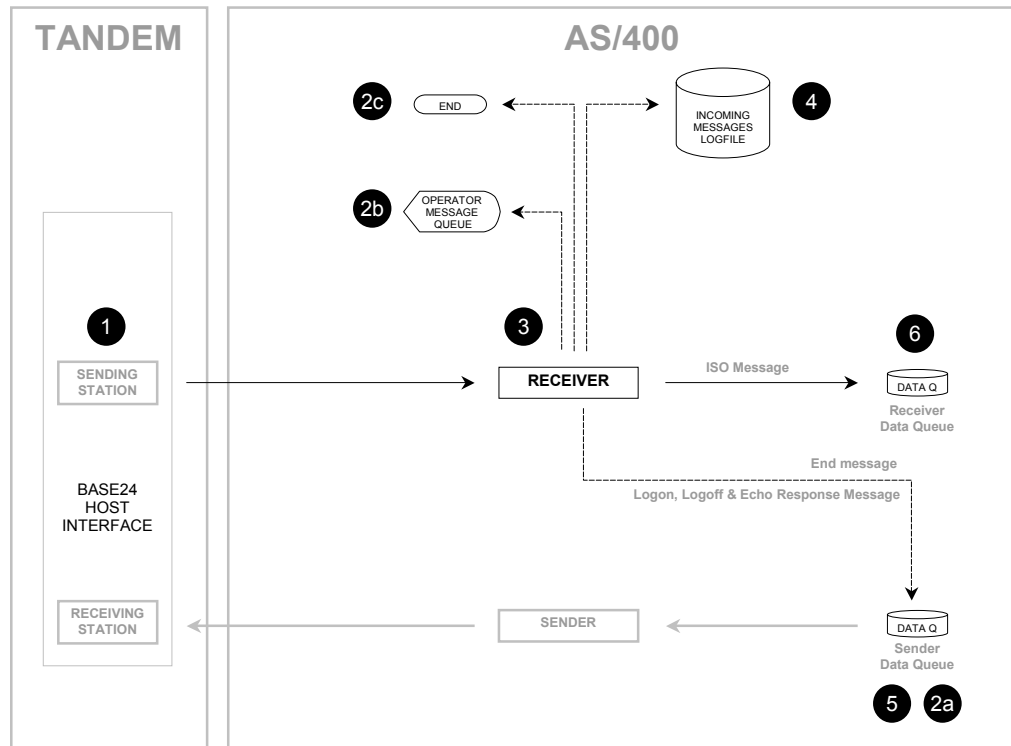
File transfer has been designed to transmit refresh and extract files between the AS/400 and TANDEM. The file transfer can only be initiated from the AS/400.

Some of the limitations of the file transfer program are:

- Only Display Text Format Characters can be transmitted. Other formats may be transmitted, but the result cannot be predicted.
- The record length of a file may not exceed 2048 bytes.

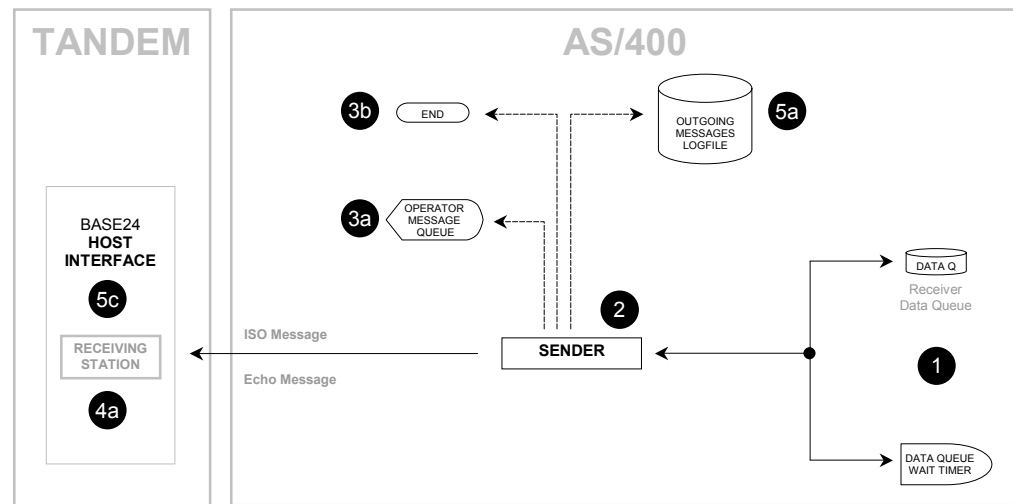
Transaction Processing

Receiver



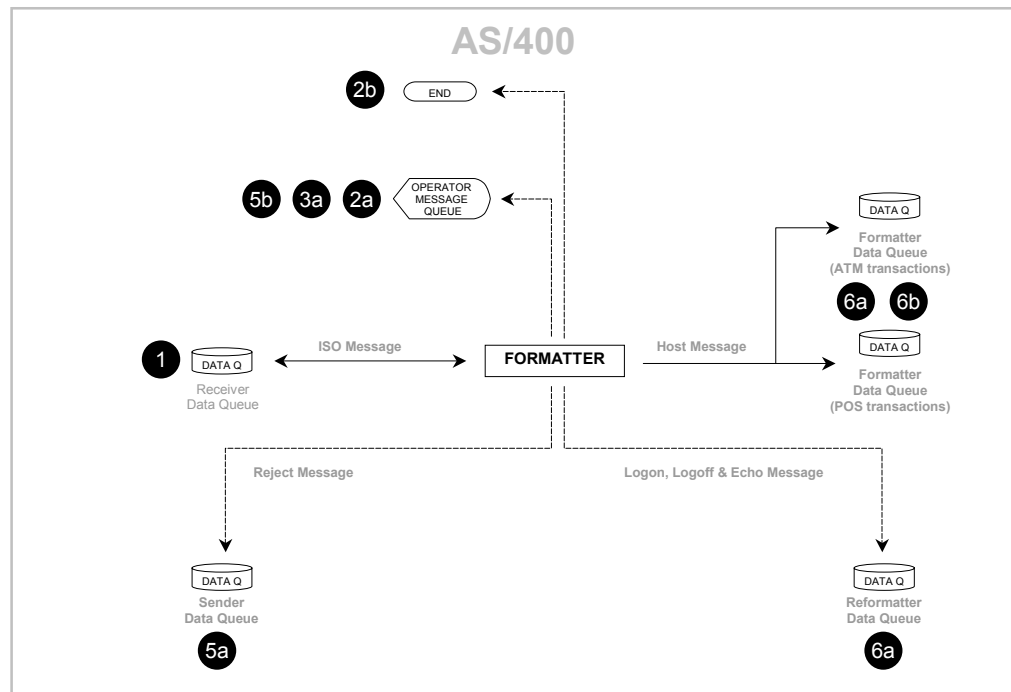
1. The Receiver waits until BASE24 sends a message.
2. If the Receiver detects an error in the conversation or the associated sending station has been stopped
 - a. It sends an *end message* to the Sender.
 - b. It sends an informational message to the operator message queue.
 - c. It ends immediately.
3. The Receiver receives a message from BASE24. If the *translate flag* is set to YES, it translates the original message's character set from ASCII to EBCDIC.
4. If the *log incoming messages flag* is set to YES, the Receiver writes the message to the incoming messages logfile.
5. If the message is an network management message
 - a. If this is a logon message and *respond to logon flag* is set to YES, the Receiver builds the response message and sends it to the Sender.
 - b. If this is a logoff message and *respond to logoff flag* is set to YES, the Receiver builds the response message and sends it to the Sender.
 - c. If this is an echo message and *respond to echo flag* is set to YES, the Receiver builds the response message and sends it to the Sender.
6. Otherwise, the Receiver sends the message to the Formatter (or host application if the Formatter is not used).
7. It waits until it receives another message from BASE24 (i.e. go back to Step #1).

Sender



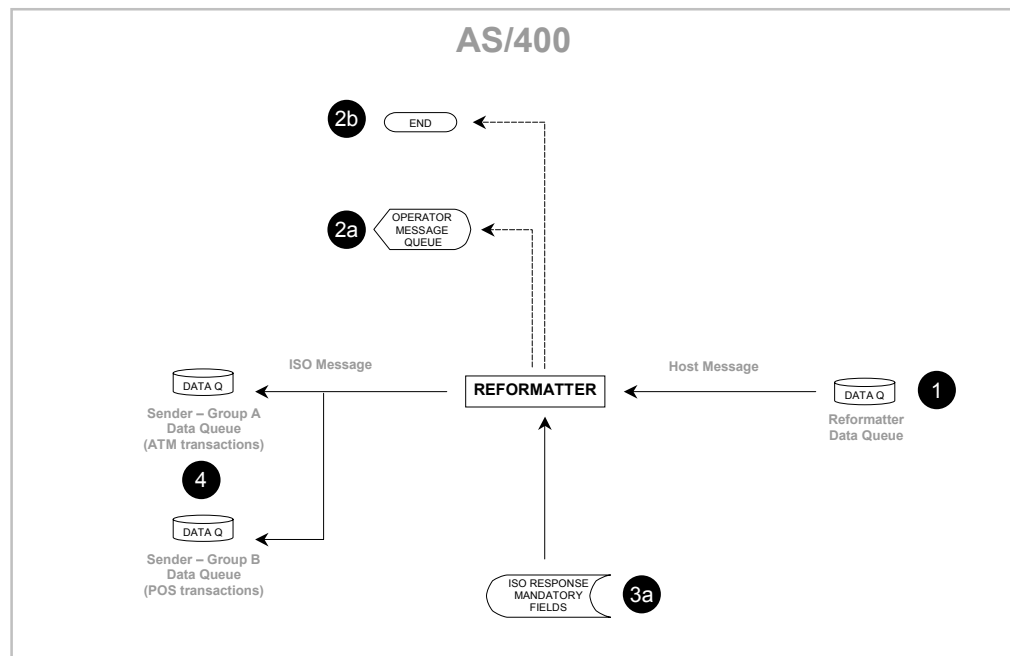
1. The Sender waits until it receives a message from the Formatter (or host application if the Formatter is not used). However, if an *echo test time interval* has been specified, the Sender will use this information as the maximum time it will wait for a message. When the wait time expires, it proceeds with its normal processing regardless of whether there is a message or not.
2. The Sender receives a message or the wait time expires.
3. If there is an end message
 - a. Sender sends an informational message to the operator message queue.
 - b. It ends immediately.
4. If there is no message
 - a. It sends an echo message to BASE24.
 - b. It waits until it receives another message or its wait timer expires (i.e. go back to Step #1).
5. Otherwise, there is a BASE24 message
 - a. If the *log outgoing messages flag* is set to YES, the Sender writes the message to the outgoing messages logfile.
 - b. It sends the message to BASE24. If the *translate flag* is set to YES, the Sender translates the message character set from EBCDIC to ASCII
6. If it detects an error in the conversation or the associated sending station has been stopped, the Sender sends an informational message to the operator and ends immediately (see Step #3).
7. It waits until it receives another message or its wait timer expires (i.e. go back to Step #1).

Formatter



1. The Formatter waits until it receives a message.
2. If the Formatter receives an *end message*
 - a. It sends an informational message to the operator message queue.
 - b. It ends immediately.
3. If it receives an invalid ISO message (i.e. “ISO” literal is missing)
 - a. It sends an informational message to the operator message queue.
 - b. It drops the message and waits for the next message (i.e. go back to Step #1).
4. Otherwise, the Formatter parses the message to its individual primary and secondary bitmap fields and builds the host message. Note that the Formatter appends the bitmap fields at the end of the host message (which will be used later by the Reformatter to rebuild the ISO message).
5. If the parsing fails
 - a. The Formatter rejects the message and sends it back to the Sender.
 - b. It sends an informational message to the operator message queue.
 - c. It waits for the next message to process (i.e. go back to Step #1).
6. Otherwise
 - a. If the message is a network management message, the Formatter sends the message to the host application or back to the Sender, depending on what is specified in the *Formatter Routing Table*.
 - b. Otherwise, the it sends the message to the host. Note that the Formatter can send all messages to one host application, or alternatively send BASE24-atm messages to one host and BASE24-pos messages to another host. Again, this depends on what is specified in the *Formatter Routing Table*.
7. The Formatter waits for the next message (i.e. go back to Step #1).

Reformatter



1. The Reformatter waits until it receives a message.
2. If the Reformatter receives an *end message*
 - a. It sends an informational message to the operator message queue.
 - b. It ends immediately.
3. Otherwise,
 - a. It loads the mandatory bitmap indicators. This lets the Reformatter determine the required bitmaps in the response message.
 - b. It rebuilds the ISO message. Note that the original bitmap fields are appended at the end of the host message so the Reformatter can still use these when it rebuilds the message.
4. The Reformatter sends the ISO-formatted message to the Sender. Note that the Reformatter can send all messages to one Sender, or alternatively send BASE24-atm messages to one Sender and BASE24-pos messages to another Sender. This depends on what is specified in the Formatter routing table
5. The Reformatter waits for the next message (i.e. go back to Step #1).

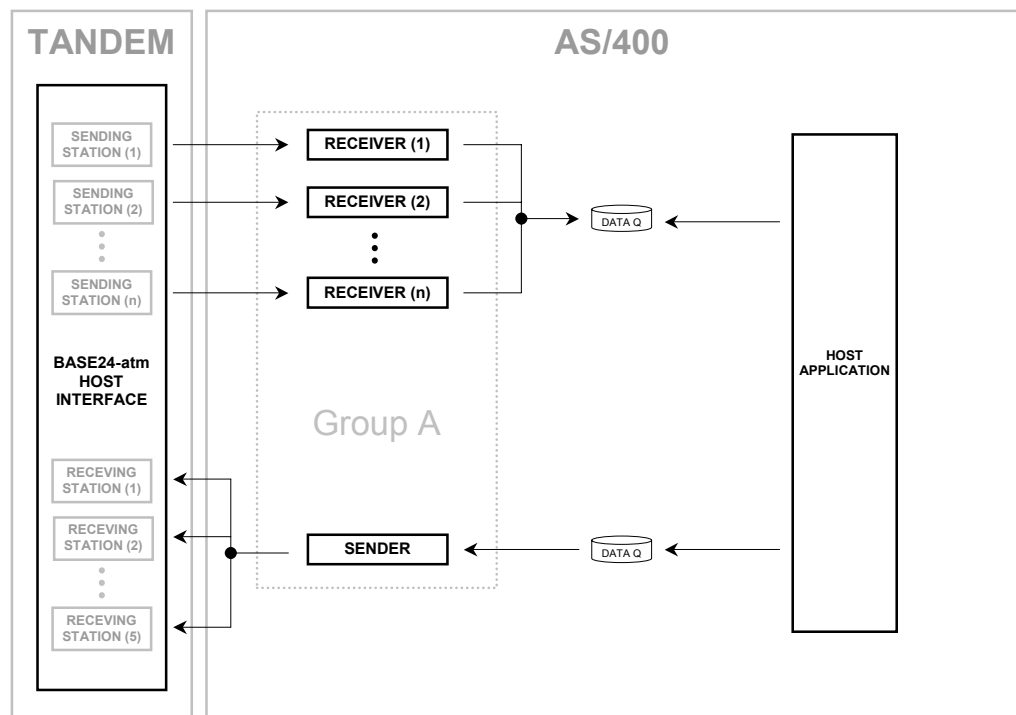
Run Configuration

Like LINK400-TCP/IP, LINK400-LU62 also allows multiple configurations to suit different requirements and environment.

Communications Programs Used

In the following configurations, it is assumed that the back-end host can process ISO messages. Hence, the formatter programs will no longer be required.

BASE24-atm Conversation

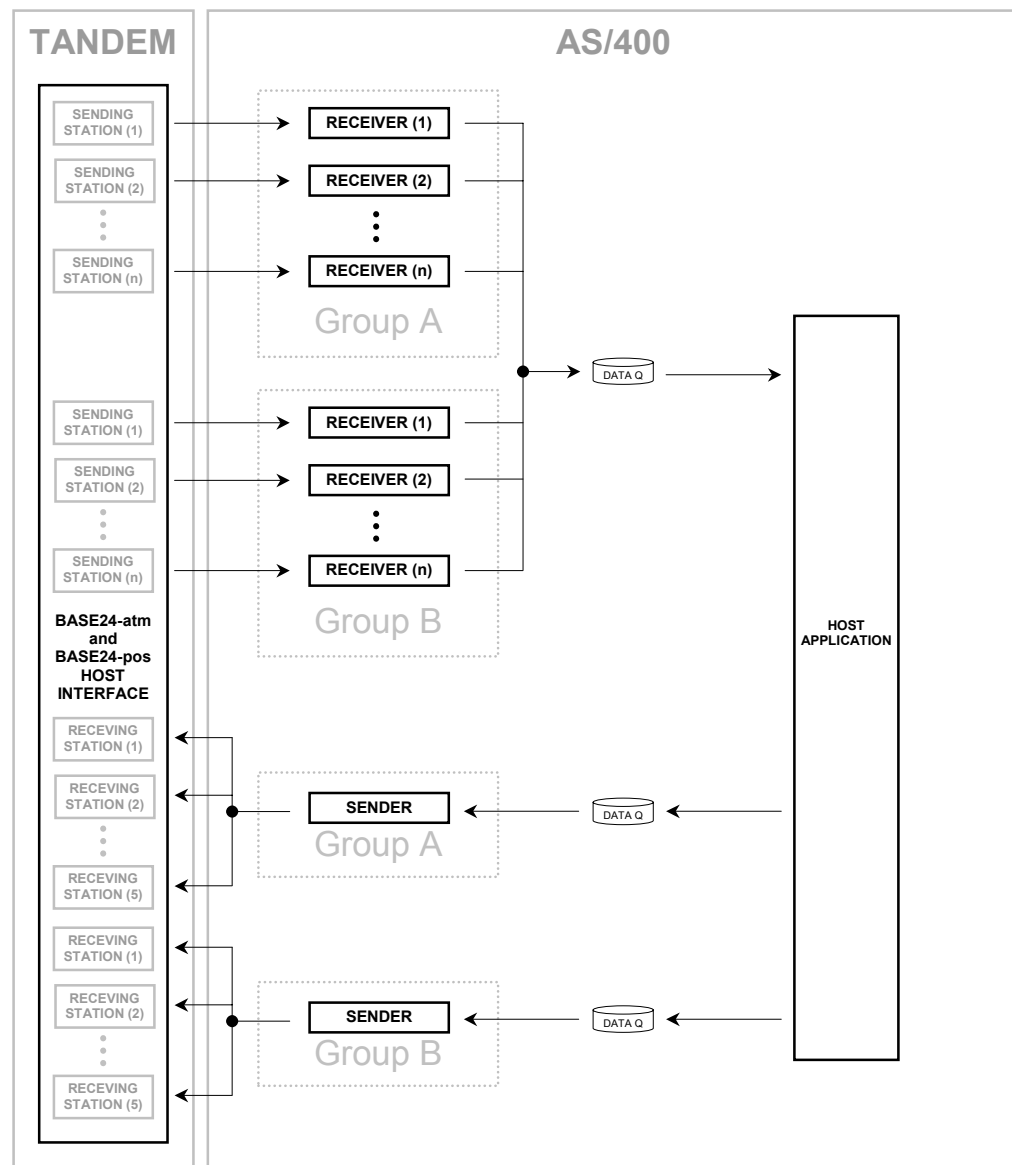


This is the simplest LINK400 configuration involving the Receiver and Sender to receive/send ATM transactions from/to BASE24. Message routers can also be increased to improve traffic throughput.

The number of Receiver conversations depends on the BASE24 and the AS/400 mode configurations. One AS/400 Receiver job will run for every Receiver conversation that is configured.

The diagram also shows that up to five Sender conversations can be set up. However, only one AS/400 Sender job that will manage all these conversations.

BASE24-atm and BASE24-pos Conversations

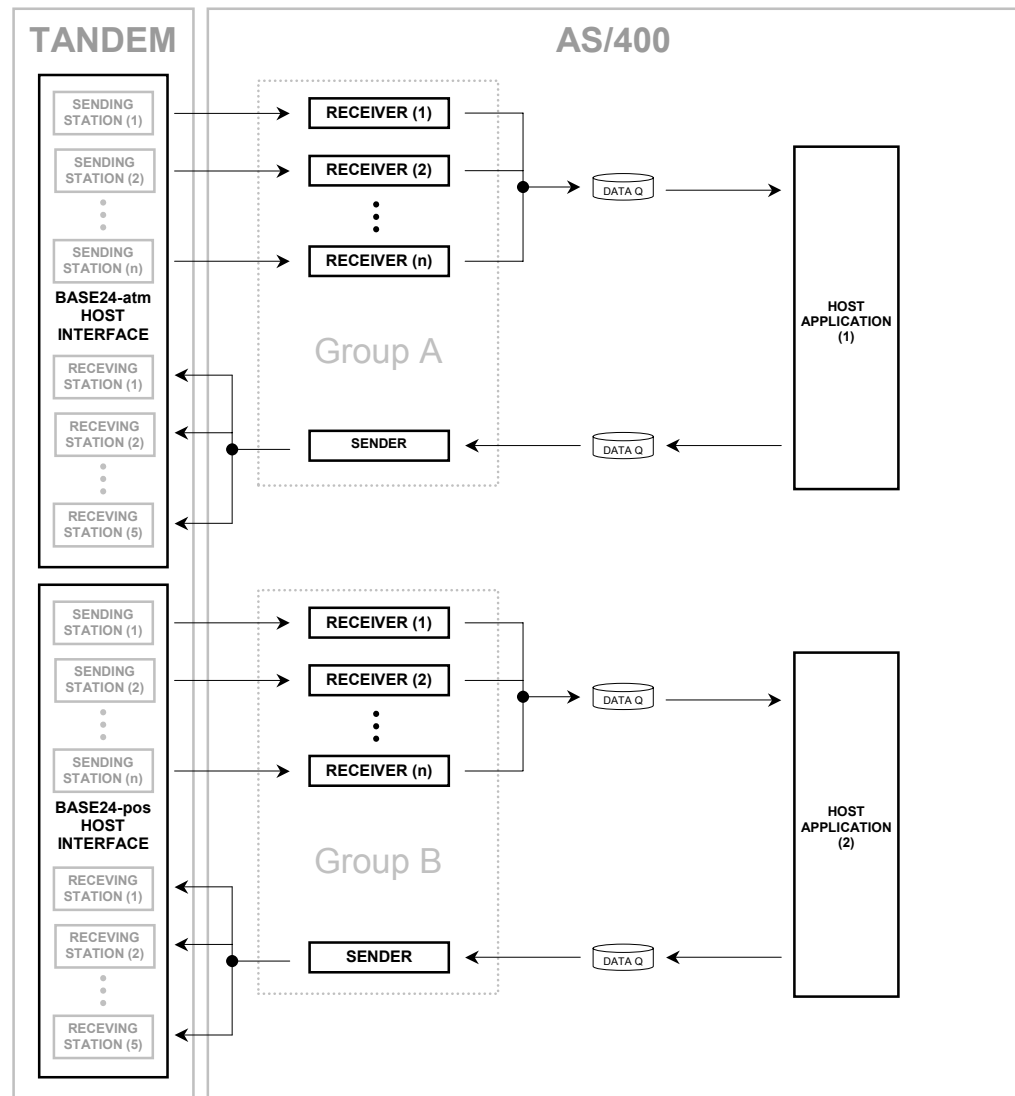


This is a typical BASE24-atm and LINK400-LU62 Receiver and Sender setup. However, BASE24-pos is added in the configuration. The Receiver and Sender processes belonging to BASE24-pos are those marked in Group B. The number of Receiver and Sender conversations that can be established with BASE24-pos is exactly similar to that of BASE24-atm.

The BASE24-pos group share the same resources that are used by the other group linked to BASE24-atm. While the same data queue can be used to send the data to a common host application, the Senders' data queues must be unique and the host application must be able to route the message to the appropriate Sender data queue.

This ensures that ATM and POS transactions are directed to BASE24-atm and BASE24-pos respectively.

BASE24-atm, BASE24-pos and 2 Host Applications Conversations



This configuration is similar to the previous diagram (see *BASE24-atm and BASE24-pos Conversations* diagram) except that ATM and POS transactions are directed to two host applications.

Here, the Receiver and Sender data queues are unique in both groups. This allows one host application to process ATM-only transactions and another host application to process POS-only transactions.

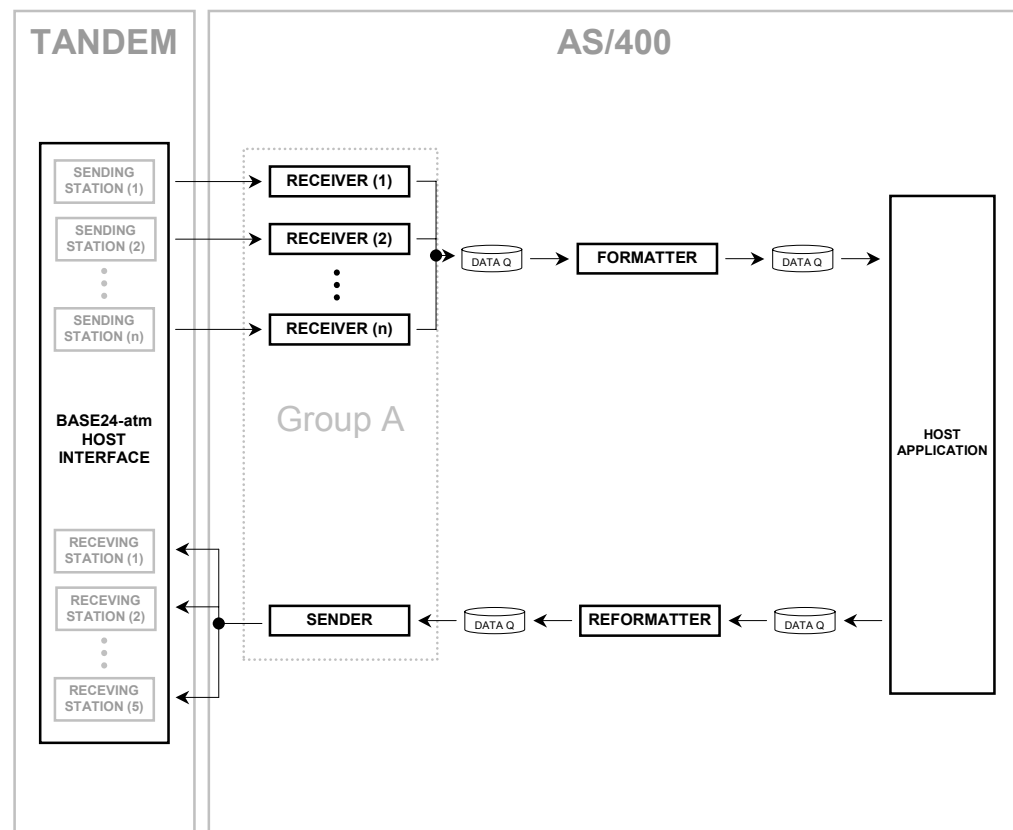
Communications and Formatter Programs Used

The Formatter and Reformatter appear when ISO messages are required to be formatted to the host application native format.

While there can be 2 Receiver and Sender groups that can be configured to separately handle ATM and POS messages, only one pair of Formatter and Reformatter can run to process these messages.

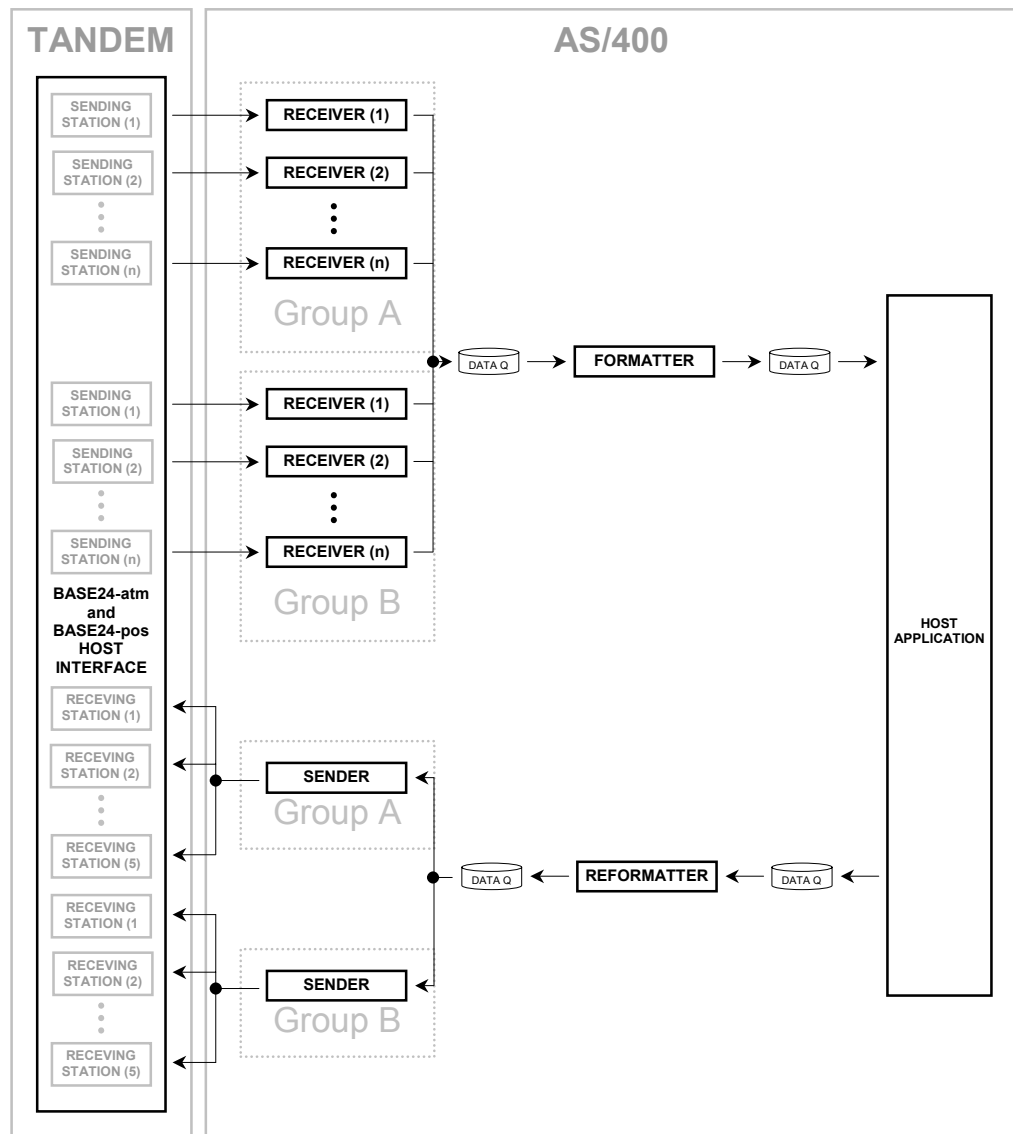
However, the Formatter can still be configured to route messages to two host applications depending on the *product ID*. Likewise, the Reformatter can also send messages back to the appropriate Sender group by also looking at the *product ID*.

BASE24-atm with Formatters



This is the same basic Receiver and Sender configuration (see *BASE24-atm Conversation* diagram) but with the Formatter and Reformatter processes to format ISO messages to the host format and vice versa. Only ATM transactions are processed in this diagram.

BASE24-atm and BASE24-pos with Formatters

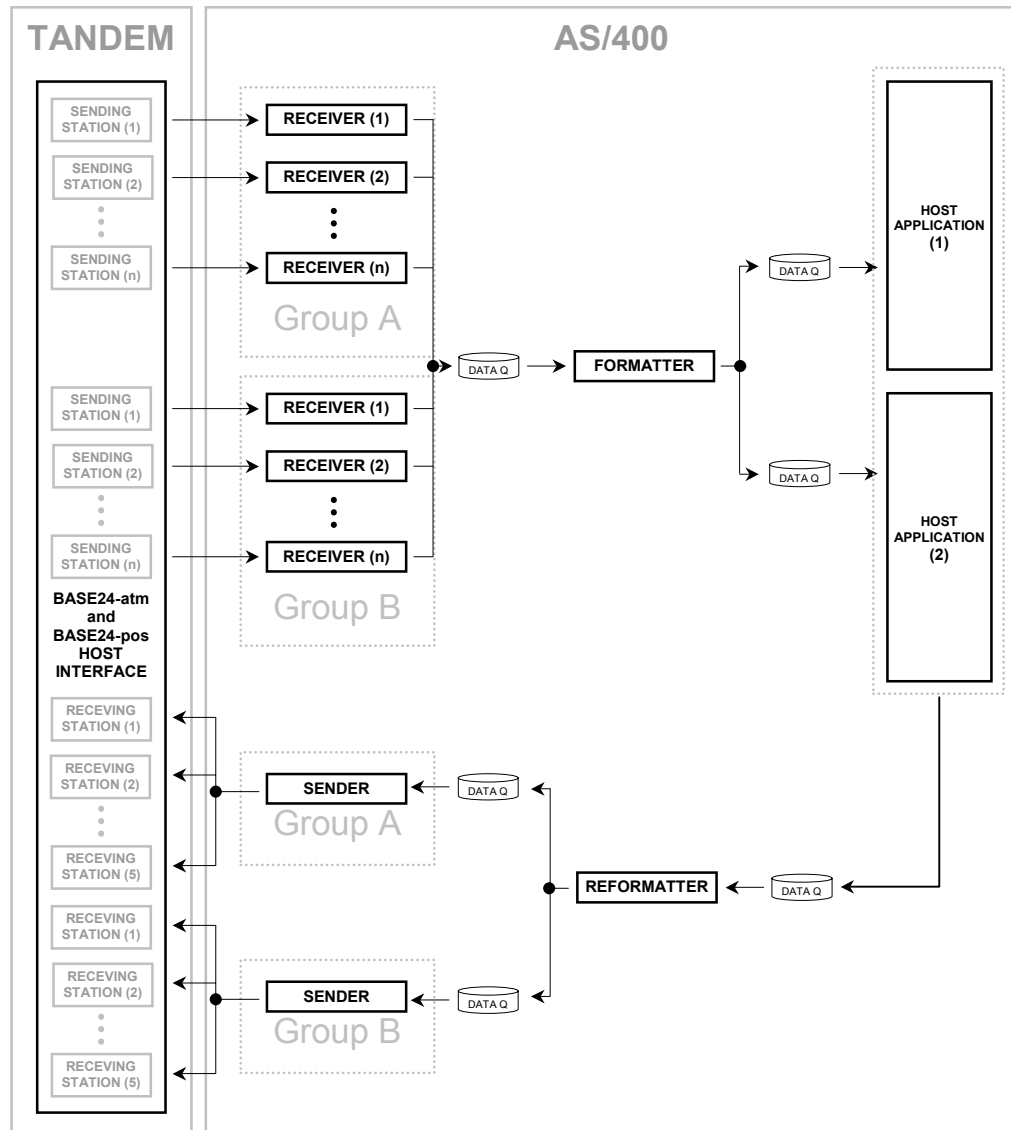


This diagram shows a normal BASE24-atm and BASE24-pos conversations involving the Formatter and Reformatter. The Formatter sends both ATM and POS transactions to one host application.

The distinct processing that must be made obvious here are:

- Since there is only one Formatter to format the messages, the Receiver for each group writes to the same data queue that the Formatter will read from.
- The Reformatter can send the message to the appropriate Sender group by checking the message product ID. This ensures that ATM and POS transactions are sent to BASE24-atm and BASE24-pos respectively.

BASE24-atm, BASE24-pos and 2 Host Applications with Formatters



This diagram shows a normal BASE24-atm and BASE24-pos conversations involving the Formatter and Reformatter and this time, two host applications.

The Formatter also has routing capabilities as the Reformatter does. It also checks the message product ID to route ATM messages to one host application and POS messages to another host application. The two host applications will send (response) messages to the same Reformatter, which knows the Sender group that the message must be routed to. The Reformatter checks the message product ID to accomplish this. Again, ATM and POS transactions are successfully sent to BASE24-atm and BASE24-pos respectively.

Section 5

Requirements and License

This section lists all the software and hardware that are required to run the TCP/IP and SNA LU6.2 versions of LINK400, including some final remarks on the LINK400 product.

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LINK400-TCP/IP Requirements

AS/400

Software

Operating System	OS/400 V3R2 and above
Compiler	ILE C/400
	CL/400
Protocol Support	TCP/IP Protocol Support

Hardware

Disk Space	12 Megabytes minimum (The size only accounts for the base software. More disk space will be required to compensate for the actual data stored in the database files and data queues. The additional disk-space size will depend on the transaction volume, the logfile retention period and the IPL frequency.)
Logical Port IDs	One port must be allocated for each Receiver and Sender conversation.

Tandem

Software

Protocol Support	XPNET TCP/IP Protocol Support
	Tandem TCP/IP Protocol Support

LINK400-LU62 Requirements

AS/400

Software

Operating System	OS/400 V3R2 and above
Compiler	RPG/400
	CL/400
Protocol Support	SNA LU6.2 Protocol Support

Hardware

Disk Space	5 Megabytes minimum (The size only accounts for the base software. More disk space will be required to compensate for the actual data stored in the database files and data queues. The additional disk-space size will depend on the transaction volume, how often they housekeep the logfiles and perform an IPL.)
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Tandem

Software

Protocol Support	XPNET SNA LU6.2 Protocol Support
	Tandem SNA LU6.2 Protocol Support

LINK400 License

Customers who use LINK400 must provide their AS/400 System Name and Serial Number to ACI who will generate the appropriate LINK400 password. The password must be updated in the B24PWD data area prior to running LINK400. LINK400 checks the license password every time it starts. If the password is invalid, it will immediately stop running.

In LINK400-TCP/IP, the Receiver source code is not provided to the customer. Additionally, in LINK400-LU62, both Receiver and Sender source codes are not given to the customer as well.